PROJECT TITLE : **Design and apply context-aware access controls based on user identity, device posture, location,and application risk**

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INTRODUCTION:

In today's digital world, organizations store a large amount of sensitive information in online systems and applications. Protecting this data from unauthorized access has become very important. Traditional access control methods mainly depend on usernames and passwords, but these methods are not always secure because attackers can steal or misuse login credentials.

To improve security, many organizations are now using context-aware access control systems. These systems make access decisions not only based on the user's identity but also on other important factors such as device posture, location, and application risk. By considering these factors, the system can decide whether a user should be allowed to access a particular resource or not

User identity helps verify who is requesting access. Device posture checks whether the device being used is secure and meets the required security standards. Location identifies where the user is accessing the system from, which helps detect suspicious access attempts. Application risk evaluates how sensitive or critical an application is before granting access.

This project focuses on designing and applying a context-aware access control mechanism that improves security by analyzing multiple conditions before allowing access. By implementing such a system, organizations can reduce security risks and ensure that only authorized users can access important data and applications.

This project focuses on designing and applying context-aware access controls by integrating different contextual factors into the access decision process. The system evaluates user identity to verify the authenticity of the requester, examines device posture to ensure that the device meets security standards, checks the geographic or network location of the request, and assesses the sensitivity or risk level of the application being accessed. Based on these factors, the system dynamically determines the appropriate level of access or security verification required.

ABSTRACT:

In modern digital environments, organizations face increasing security challenges due to the widespread use of cloud services, remote work, and mobile devices. Traditional access control systems that rely only on usernames and passwords are no longer sufficient to protect sensitive data and applications. As cyber threats become more sophisticated, there is a growing need for smarter and more adaptive security mechanisms that can evaluate multiple factors before granting access.

This project focuses on designing and implementing **context-aware access controls** that dynamically regulate access based on several contextual factors such as user identity, device posture, location, and application risk level. By analyzing these parameters, the system can make informed decisions about whether to allow, restrict, or deny access to resources. This approach aligns with modern security models like Zero Trust Architecture, which emphasizes continuous verification rather than assuming trust based on network location.

The proposed system evaluates the identity of the user to ensure authentication, checks the security status of the device to confirm compliance with security policies, analyzes the geographical or network location of the request, and assesses the sensitivity of the application being accessed. Based on these factors, adaptive security policies are applied, such as multi-factor authentication, restricted access, or complete denial of access in high-risk situations.

By implementing context-aware access control, the system enhances overall security while maintaining flexibility and usability for legitimate users. The project demonstrates how integrating multiple contextual attributes into access control decisions can significantly reduce unauthorized access, improve risk management, and strengthen the protection of organizational resources in modern computing environments.

As organizations increasingly depend on digital platforms and cloud-based applications, protecting sensitive information from unauthorized access has become a critical concern. Traditional security models that rely on perimeter-based defenses are no longer effective in modern distributed environments. This has led to the development of advanced security approaches such as context-aware access control.

Use Case Scenarios

Scenario 1: Finance Staff Access to ERP

A finance employee needs access to the company's ERP system to manage payroll. The user connects remotely using a company laptop and authenticates through the ZPA Client Connector with Azure AD and MFA. ZPA verifies the user's identity, confirms membership in the **Finance_ERP_Users** group, and checks device security such as antivirus and encryption. After verification, ZPA creates a secure micro-tunnel only to the ERP application.

Output:

• User authenticated and authorized
• Secure micro-tunnel created for ERP access
• Other internal systems remain inaccessible

Scenario 2: Contractor Access to Git Repository

A temporary contractor requires access to a specific Git repository for a project. The contractor is assigned to the **Project_X_Contractors** group in the identity system. When logging in through ZPA, the system verifies identity and checks device compliance such as firewall and OS updates. Access is granted only to the Git repository during business hours.

Output:

• Role-based and time-restricted access provided
• Access limited to the Git repository
• Other internal resources blocked

Scenario 3: Executive Access to Sensitive Documents

A company executive needs to access confidential financial documents stored on SharePoint while traveling. The user logs in through ZPA using a company tablet and MFA authentication. ZPA verifies the executive role and device compliance but detects login from a foreign location. Access is allowed with restrictions such as disabled downloads and monitored activity.

Output:

• User authenticated but location flagged
• Restricted read-only access granted
• Security team notified of high-risk login

Scenario 4: IT Technician Remote Support Access

An IT technician connects from home to provide remote support. The technician logs in through ZPA with Azure AD and MFA. ZPA verifies membership in the **IT_Support** group and confirms the device is compliant. Secure micro-tunnels are created to access ServiceNow and the remote desktop support server.

Output:

• Technician authenticated and authorized
• Secure access to support tools only
• Other internal systems remain hidden

Scenario 5: CRM Access from Public Wi-Fi

A marketing employee needs to update campaign details in the Salesforce CRM while connected to public Wi-Fi. The user logs in through ZPA using MFA. Although the user and device are trusted, the public network is considered risky. ZPA routes the session through a secure browser isolation environment to protect sensitive data.

Output:

- User authenticated but network flagged as risky
- Access provided through browser isolation
- Data protection controls enforced

Stage – 1

Title:

Design and apply context-aware access controls based on user identity, device posture, location, and application risk.

Overview:

In response to the evolving threat landscape and the dissolution of the traditional network perimeter, modern enterprises are rapidly shifting away from legacy VPN-centric remote access models. These conventional approaches often grant excessive network-level access, creating a significant attack surface and increasing the risk of lateral threat movement. To address these challenges, this project centers on the implementation of a Zero Trust security architecture, fundamentally operating on the principle of "never trust, always verify." By moving beyond location-based access, we aim to establish a security posture where trust is no longer implicit, and access is determined by granular context rather than network connectivity.

The core of this initiative involves the deployment of Zscaler Private Access (ZPA), a cloud-delivered Zero Trust Access solution designed to create identity-aware, application-specific connectivity. Unlike traditional VPNs that place users on the internal network, ZPA leverages a "service-initiated" model to establish direct, one-to-one connections between authorized users and specific applications, completely hiding the corporate network and IP address space from the internet. This ensures that users connect exclusively to the applications they are permitted to access, based on continuous identity verification and context, rendering the applications invisible to unauthorized users and preventing common attack vectors such as port scanning and distributed denial-of-service (DDoS) attacks.

Ultimately, this project will enforce a least-privilege access model, ensuring users and devices are granted the minimum required access necessary to perform their functions. By decoupling application access from network access, we will eliminate the risk of lateral movement across the network, as users have no visibility or connectivity to other hosts. This strategic move not only significantly enhances security by reducing the attack surface but also simplifies the user experience and reduces operational overhead associated with legacy remote access infrastructure, positioning the organization at the forefront of secure and efficient access management.

List of Vulnerability:

S. No.	Vulnerability Name	CWE No.
1.	Absolute Path Traversal	CWE 36
2.	Sensitive Cookie in HTTPS Session Without 'Secure' Attribute	641
3.	SQL Injection	89
4.	Broken Authentication	285
5.	Insecure Direct Object Reference	639
6.	Security Misconfiguration	732
7.	Cross-Site Request Forgery	352
8.	Cleartext Transmission of Sensitive Information	319
9.	ClickJacking	1021
10.	webserver allows password auto-completion	310

Report:

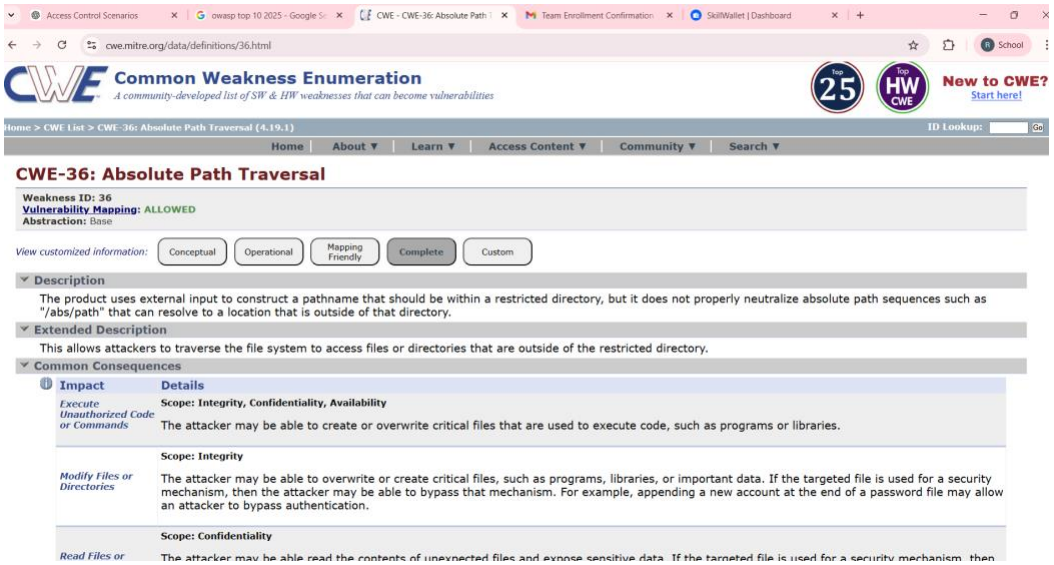
1. Vulnerability name: Absolute Path Traversal

2. CWE: 36

OWASP category: A01:2025 Broken Access Control

Description:Absolute Path Traversal occurs when a web application allows users to access files on the server without proper validation of the file path. Attackers can manipulate input parameters to access sensitive system files outside the intended directory.

Business Impact: This vulnerability may allow attackers to read sensitive files such as configuration files, system files, or application data. It can lead to information disclosure and further system compromise..

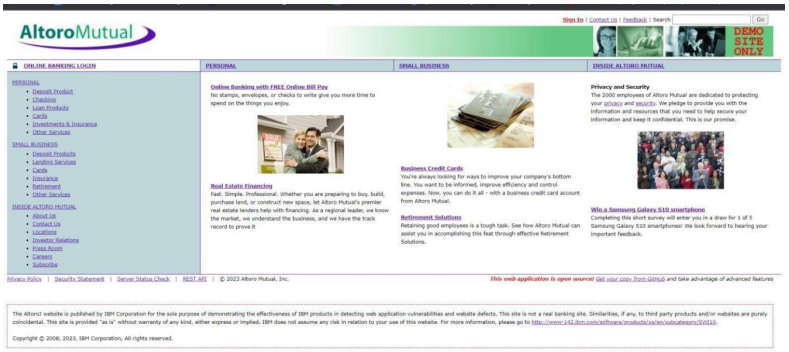


Vulnerability path: http://testfire.net/

Vulnerability parameter: http://testfire.net/showfile.jsp?file=../../../../etc/passwd

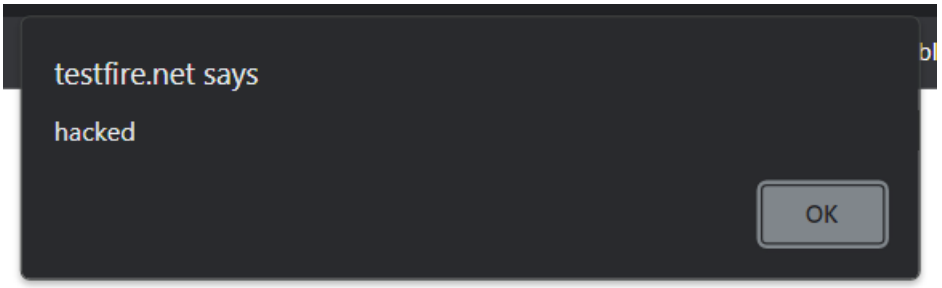
Steps to Reproduce:

- Access the URL



- In the search box we will input some code to perform the vulnerability.

- The Script we will be inputting is '><script>alert('hacked')</script>
- This displays a harmless pop up alert box with the text saying 'hacked'



Recommendation:

- When interacting with databases, use parameterized statements or prepared statements to avoid SQL injection, which can be a vector for XSS.
- Sanitize and validate all user inputs and ensure that any data displayed on the web page is properly encoded. Use output encoding libraries or functions to prevent script injection.

3. Vulnerability name: Broken access control

CWE: 285

Description: When access control checks are not applied consistently - or not at all - users are able to access data or perform actions that they should not be allowed to perform. This can lead to a wide range of problems, including information exposures, denial of service, and arbitrary code execution.

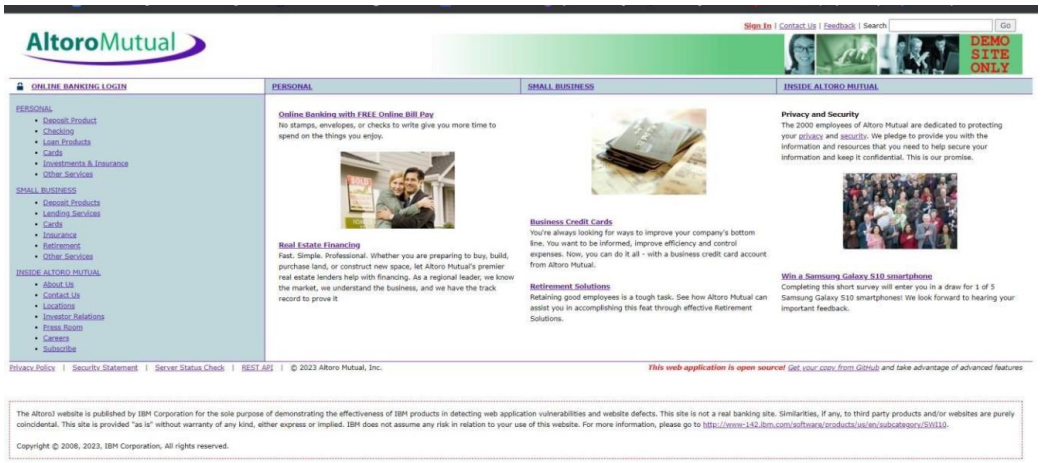
Business Impact: wide-ranging business impact, including financial losses, reputation damage, legal consequences, operational disruption, and loss of customer trust. Addressing this vulnerability is crucial to protect the organization and its stakeholders from these potential negative effects.

Vulnerability path: http://testfire.net/

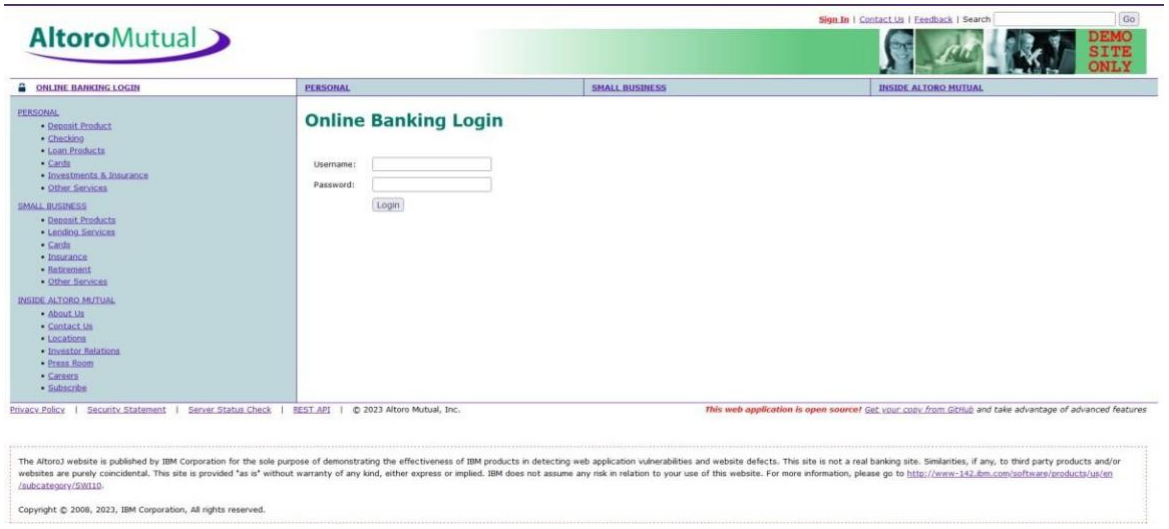
Vulnerability parameter: http://testfire.net/login.jsp

Steps to Reproduce:

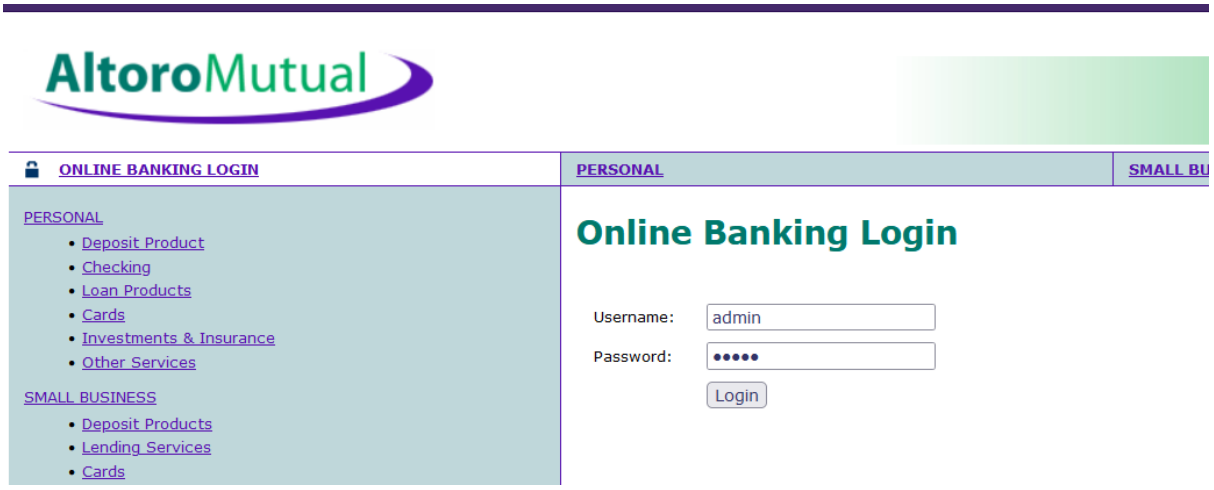
- Access the URL



- Now we will try to sign in to this website with admin privileges



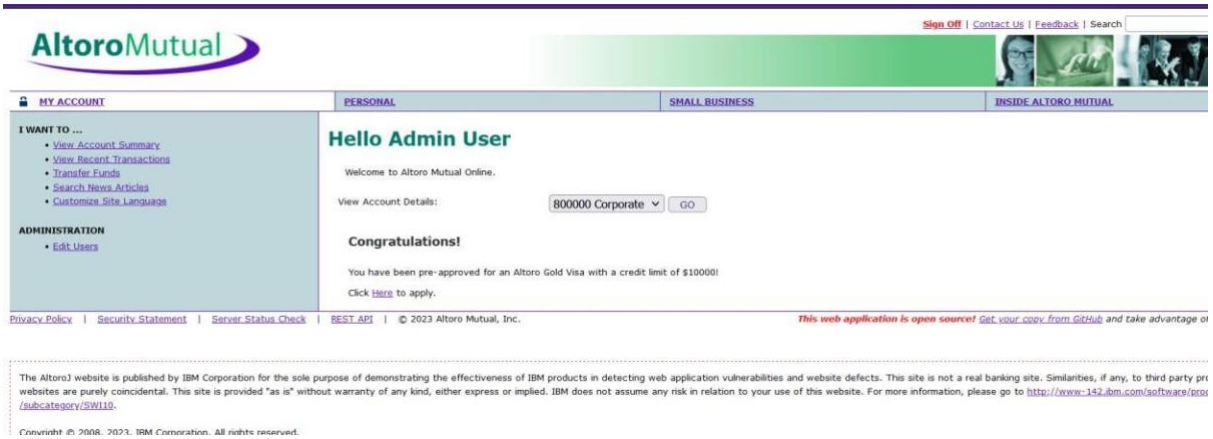
- Also, we will be using burp suite to get requests from the website and know additional information. We use 'admin' for the username and password.



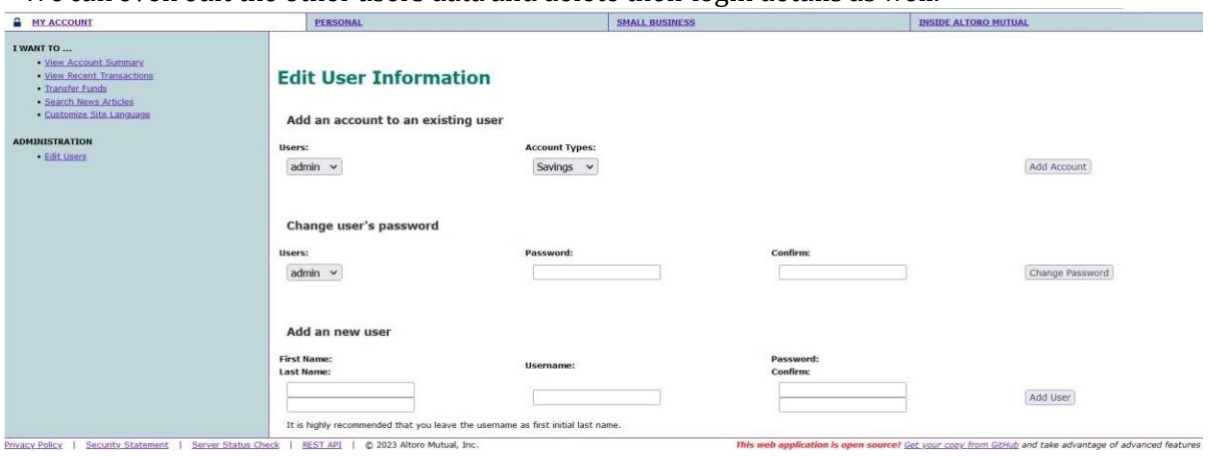
- This request has been received in the burp suite with the username and password as well in clear text.

```
Pretty   Raw   Hex
1  POST /doLogin HTTP/1.1
2  Host: testfire.net
3  User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64; rv:109.0) Gecko/20100101 Firefox/117.0
4  Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,*/*;q=0.8
5  Accept-Language: en-US,en;q=0.5
6  Accept-Encoding: gzip, deflate, br
7  Content-Type: application/x-www-form-urlencoded
8  Content-Length: 37
9  Origin: http://testfire.net
10 Connection: close
11 Referer: http://testfire.net/login.jsp
12 Cookie: JSESSIONID=B6ADB36ACD5C83083787343A1F97F853; AltoroAccounts="ODAwMDAwfkNvcnBvcmFOZX45LjQ30TA1MTE2MUU3fDgwMDAwMX5DaGVja2luZ34tNC4yMjc0NjkONTZFN3w="
13 Upgrade-Insecure-Requests: 1
14
15 uid=admin&passw=admin&btnSubmit=Login
```

- now we click on forward request in the burp suite and then we will be redirected to the admin user details.



- We can even edit the other users data and delete their login details as well.



Recommendations:

- Ensure that the application has a robust authentication and authorization mechanism in place. Users should be authenticated before any sensitive actions are performed, and authorization checks should be conducted to verify that users have the necessary permissions for the requested operation.
- Ensure that session management is secure. Use strong session IDs, enforce session timeouts, and regenerate session tokens after login. Implement session fixation protection to prevent session hijacking.

4. Vulnerability name: SQL injection

   CWE: 89

   Description: The product constructs all or part of an SQL command using externally-influenced input from an upstream component, but it does not neutralize or incorrectly neutralizes special elements that could modify the intended SQL command when it is sent to a downstream component.

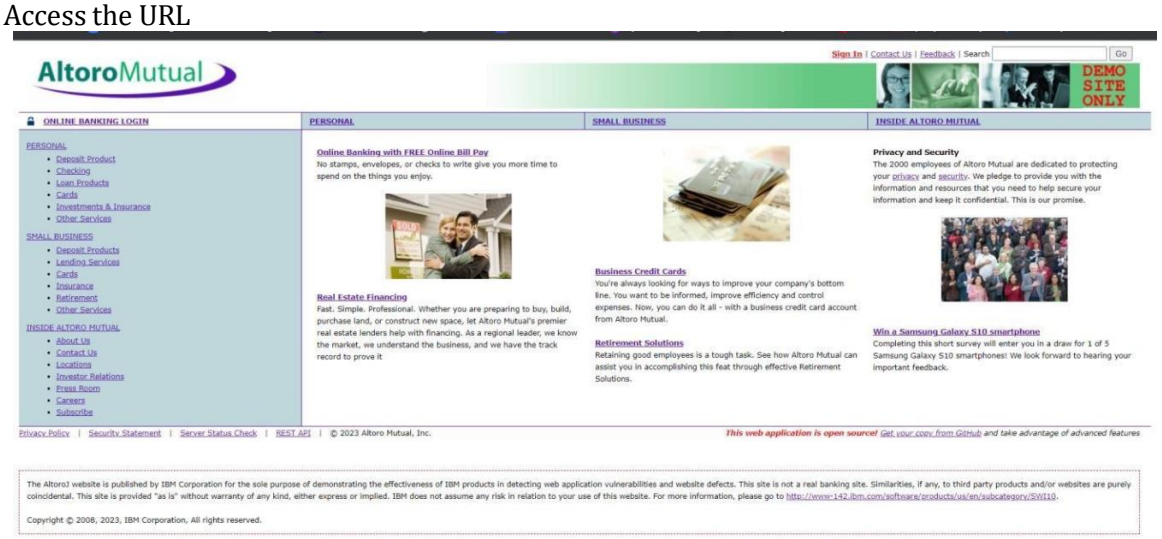
Business Impact: In summary, it is crucial to underscore that CWE-89, known as SQL Injection, can exert a profound and diverse business impact. This encompasses critical facets such as data breaches, financial setbacks, harm to reputation, legal ramifications, and operational turmoil. Hence, the imperative of preventing and remedying SQL injection vulnerabilities cannot be overstated, as it is indispensable for fortifying the security and continuity of an organization's applications and data.

Vulnerability path: http://testfire.net/

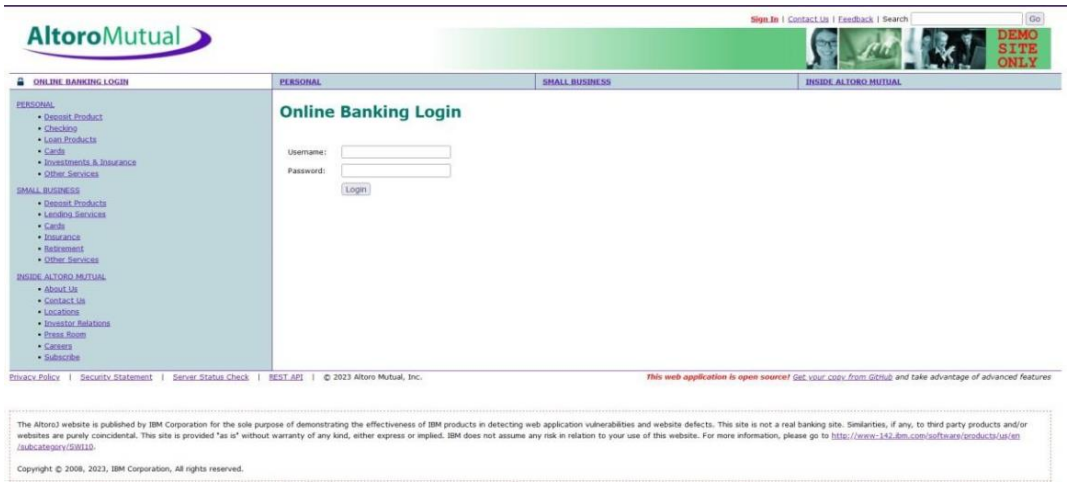
Vulnerability parameter: http://testfire.net/login.jsp

Steps to Reproduce:

- Access the URL



- Now we will try to sign in to this website with admin privileges but using SQL injection



- Also, we will be using burp suite to get requests from the website and know additional information.

Intercept | HTTP history | WebSockets history | Proxy settings

Request to http://testfire.net:80 [65.61.137.117]

Forward | Drop | Intercept is on | Action | Open browser

Pretty | Raw | Hex

```
1 GET / HTTP/1.1
2 Host: testfire.net
3 User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64; rv:109.0) Gecko/20100101 Firefox/118.0
4 Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,*/*;q=0.8
5 Accept-Language: en-US,en;q=0.5
6 Accept-Encoding: gzip, deflate, br
7 Connection: close
8 Cookie: JSESSIONID=AFA9012F50640AF57ED2D63F8959DEE0
9 Upgrade-Insecure-Requests: 1
10
11
```

Dashboard | Target | Proxy | Intruder | Repeater | Collaborator | Sequencer | Decoder | Comparer | Logger | Organizer | Extensions | Learn

Intercept | HTTP history | WebSockets history | Proxy settings

Request to http://testfire.net:80 [65.61.137.117]

Forward | Drop | Intercept is on | Action | Open browser

Pretty | Raw | Hex

```
1 POST /doLogin HTTP/1.1
2 Host: testfire.net
3 User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64; rv:109.0) Gecko/20100101 Firefox/118.0
4 Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,*/*;q=0.8
5 Accept-Language: en-US,en;q=0.5
6 Accept-Encoding: gzip, deflate, br
7 Content-Type: application/x-www-form-urlencoded
8 Content-Length: 45
9 Origin: http://testfire.net
10 Connection: close
11 Referer: http://testfire.net/login.jsp
12 Cookie: JSESSIONID=AFA9012F50640AF57ED2D63F8959DEE0
13 Upgrade-Insecure-Requests: 1
14
15 uid=%27+or+1%3D1--&passw=1234&btnSubmit=Login
```

- Click forward request multiple times to proceed to the login page.

AltoroMutual

MY ACCOUNT | PERSONAL | SMALL BUSINESS | INSIDE ALTORO MUTUAL

I WANT TO ...
- View Account Summary
- View Recent Transactions
- Transfer Funds
- Search News Articles
- Customize Site Language

ADMINISTRATION
- Edit Users

Hello Admin User

Welcome to Altoro Mutual Online.

View Account Details: 800000 Corporate GO

Congratulations!

You have been pre-approved for an Altoro Gold Visa with a credit limit of $10000!

Click Here to apply.

Privacy Policy | Security Statement | Server Status Check | REST API | © 2023 Altoro Mutual, Inc. This web application is open source! Get your copy from GitHub and take advantage o

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- With this we can know that sql injection worked and we got the admin privileges.

Recommendations:

- If parameterized statements are not feasible, use proper input validation and escaping mechanisms to sanitize user inputs before they are used in SQL queries. This helps prevent malicious code injection.
- Utilize stored procedures for database operations. This can help separate SQL code from application code and reduce the risk of SQL injection.
- Implement strict input validation for all user inputs, ensuring that data adheres to the expected format and structure.

5. Vulnerability name: Broken authentication

CWE: 285

Description: The system's authorization functionality does not prevent one user from gaining access to another user's data or record by modifying the key value identifying the data.

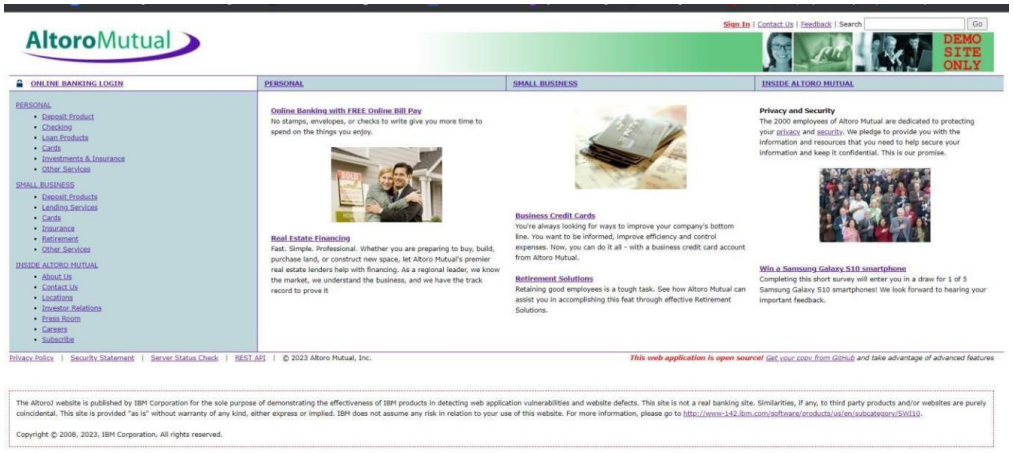
Business Impact: To effectively mitigate the business impact of CWE-285, it is imperative that organizations place a strong emphasis on bolstering their authentication and session management practices. This should encompass the adoption of multi-factor authentication, the secure storage of credentials, meticulous session handling, and a commitment to conducting routine security assessments and testing. The rectification of these vulnerabilities stands as a critical imperative, safeguarding sensitive data, user identities, and the organization's overarching security stature and reputation.

Vulnerability path: http://testfire.net/

Vulnerability parameter: http://testfire.net/login.jsp

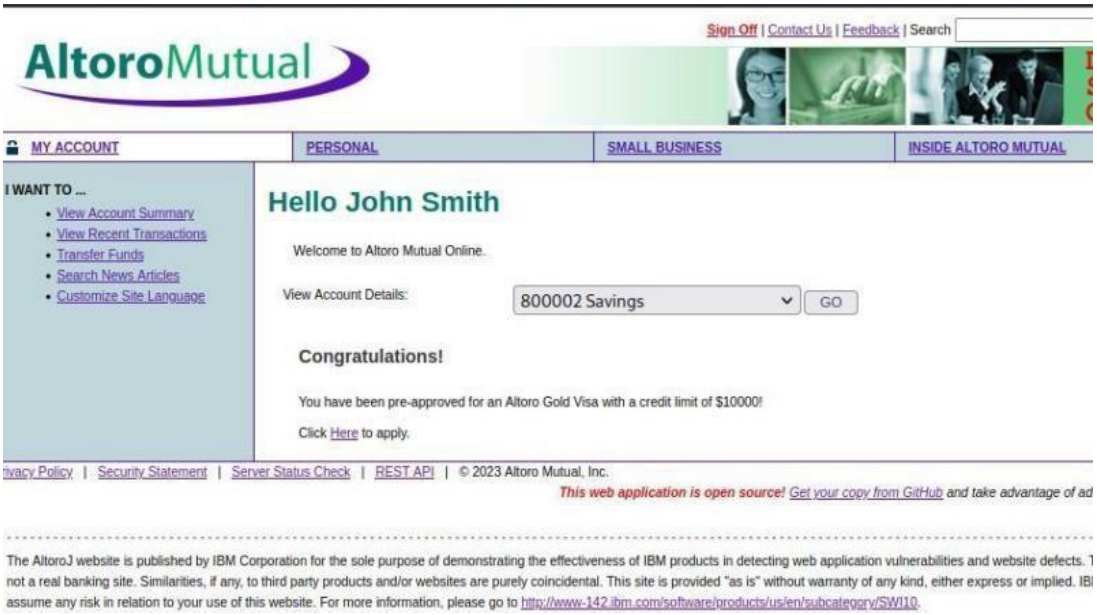
Steps to Reproduce:

- Access the URL



- Now we will try to login using some different approach.

- As we know which users are present in the database of this website by using the admin privileges. We can directly access a particular user by simply knowing their username; we will add some characters after his user name as a sql injection to simply bypass the password.



- This leads us to the details of this person's account.

Recommendations:

- Implement strong authentication mechanisms, such as multi-factor authentication (MFA) or two-factor authentication (2FA), to ensure that only authorized users can access the system.
- Implement proper authorization controls to ensure that users have the necessary permissions to access specific functions or data. Use role-based access control (RBAC) to manage user privileges effectively.
- Deploy intrusion detection and prevention systems (IDS/IPS) to monitor and block suspicious activities related to authentication and authorization.

6. Vulnerability Name: Insecure Direct object Reference

CWE: 639

OWASP Category: A01: Broken Access Control

Description:
Insecure Direct Object Reference (IDOR) is a vulnerability that arises when attackers can access or modify objects by manipulating identifiers used in a web application's URLs or parameters. It occurs due to missing access control checks, which fail to verify whether a user should be allowed to access specific data.

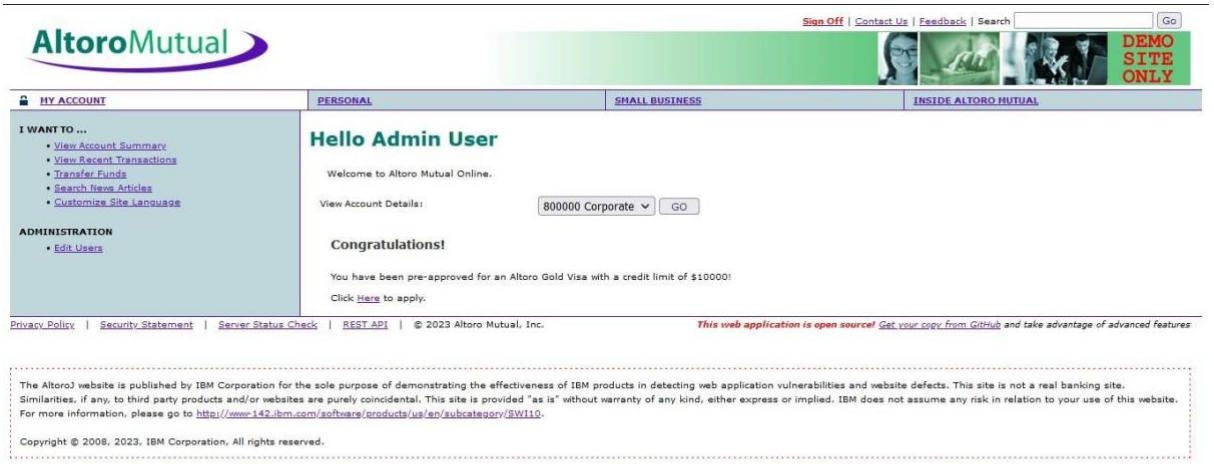
Business Impact:
To effectively mitigate the business impact of CWE-639, organizations must make it a top priority to fortify their access control and authorization mechanisms. This entails the implementation of robust security measures, the regular conduct of comprehensive security assessments, and the deployment of intrusion detection systems to promptly identify and counter unauthorized access attempts. These measures stand as absolutely critical in the defense of sensitive data, the overall security of systems, and the preservation of the organization's esteemed reputation.

Vulnerability Path: http://testfire.net

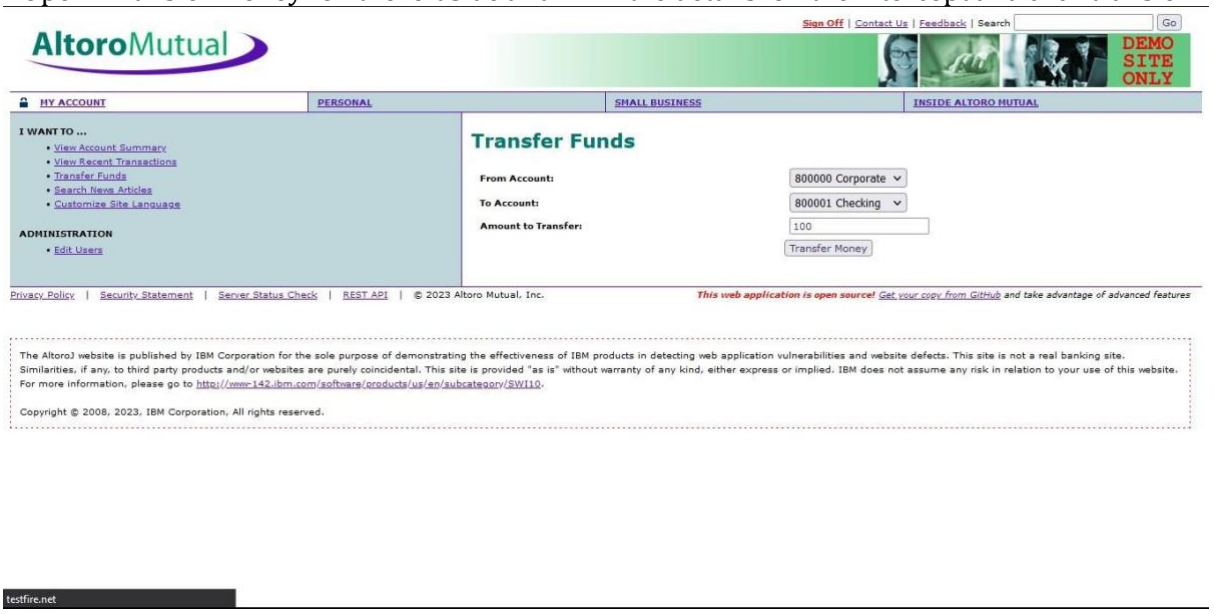
Vulnerability Parameter: http://testfire.net/bank/transfer.jsp

Steps to Reproduce:

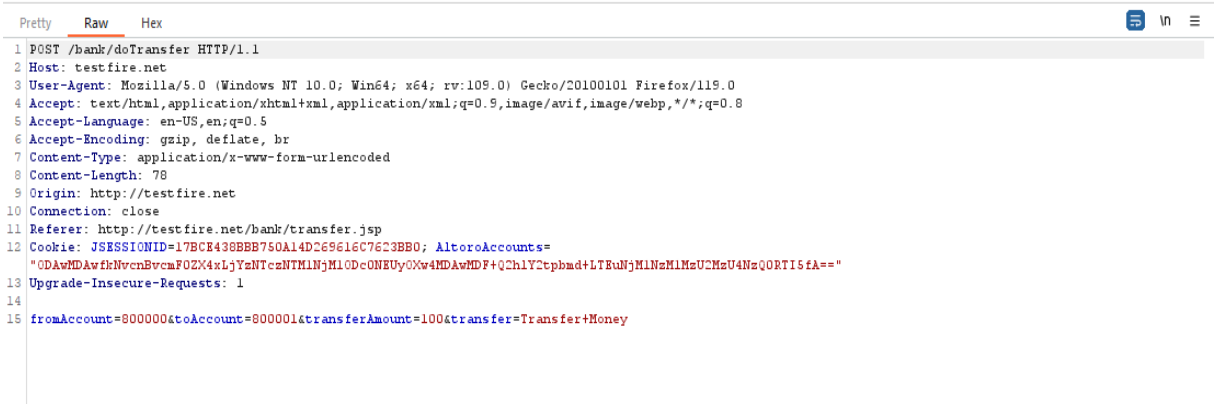
- Go to the domain.
- Sign into the site.



- Open "Transfer Money" on the left side and fill in the details. On the intercept and click transfer



- Open burp and notice the change.



- In the 15th line change the amount from 100 to 1000 and click forward

```
fromAccount=800000&toAccount=800001&transferAmount=100&transfer=Transfer+Money

fromAccount=800000&toAccount=800001&transferAmount=1000&transfer=Transfer+Money
```

Pretty Raw Hex

```
1 GET /v1/tiles HTTP/1.1
2 Host: contile.services.mozilla.com
3 User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64; rv:109.0) Gecko/20100101 Firefox/119.0
4 Accept: */*
5 Accept-Language: en-US,en;q=0.5
6 Accept-Encoding: gzip, deflate, br
7 Te: trailers
8 Connection: close
9
10
```

- Look at the site, we can notice the msg that shows the transfer of 1000.

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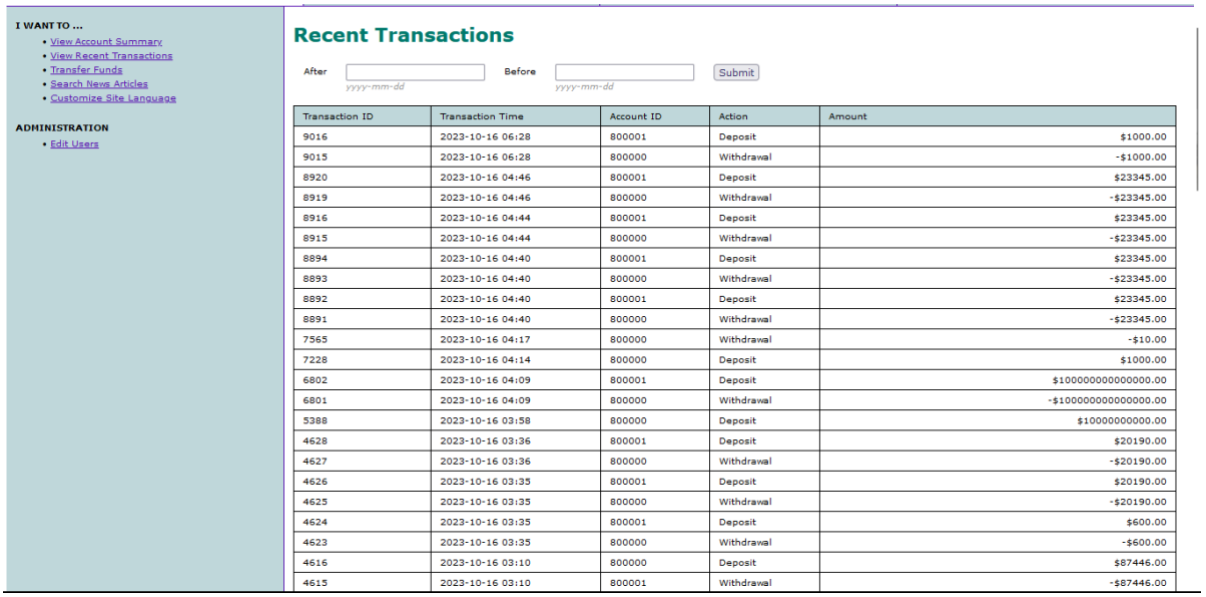
- Off the Intercept and open "View Recent Transactions".



Transaction ID	Transaction Time	Account ID	Action	Amount
9016	2023-10-16 06:28	800001	Deposit	$1000.00
9015	2023-10-16 06:28	800000	Withdrawal	-$1000.00
8920	2023-10-16 04:46	800001	Deposit	$23345.00
8919	2023-10-16 04:46	800000	Withdrawal	-$23345.00
8916	2023-10-16 04:44	800001	Deposit	$23345.00
8915	2023-10-16 04:44	800000	Withdrawal	-$23345.00
8894	2023-10-16 04:40	800001	Deposit	$23345.00
8893	2023-10-16 04:40	800000	Withdrawal	-$23345.00
8892	2023-10-16 04:40	800001	Deposit	$23345.00
8891	2023-10-16 04:40	800000	Withdrawal	-$23345.00
7565	2023-10-16 04:17	800000	Withdrawal	-$10.00
7228	2023-10-16 04:14	800000	Deposit	$1000.00
6802	2023-10-16 04:09	800001	Deposit	$100000000000000.00
6801	2023-10-16 04:09	800000	Withdrawal	-$100000000000000.00
5388	2023-10-16 03:58	800000	Deposit	$10000000000.00
4628	2023-10-16 03:36	800001	Deposit	$20190.00
4627	2023-10-16 03:36	800000	Withdrawal	-$20190.00
4626	2023-10-16 03:35	800001	Deposit	$20190.00
4625	2023-10-16 03:35	800000	Withdrawal	-$20190.00
4624	2023-10-16 03:35	800001	Deposit	$600.00
4623	2023-10-16 03:35	800000	Withdrawal	-$600.00
4616	2023-10-16 03:10	800000	Deposit	$87446.00
4615	2023-10-16 03:10	800001	Withdrawal	-$87446.00

Recommendation:

- Utilize parameterized queries or prepared statements when interacting with databases. This prevents user-controlled input from being executed as SQL code.
- Implement a Content Security Policy to mitigate the risk of cross-site scripting (XSS) attacks, which can be used to execute unauthorized queries.
- Implement account lockout mechanisms that temporarily suspend accounts after a specified number of unauthorized access attempts to prevent brute force attacks.

7. Vulnerability name: Security Misconfiguration

CWE:732

Description: The product specifies permissions for a security-critical resource in a way that allows that resource to be read or modified by unintended actors.

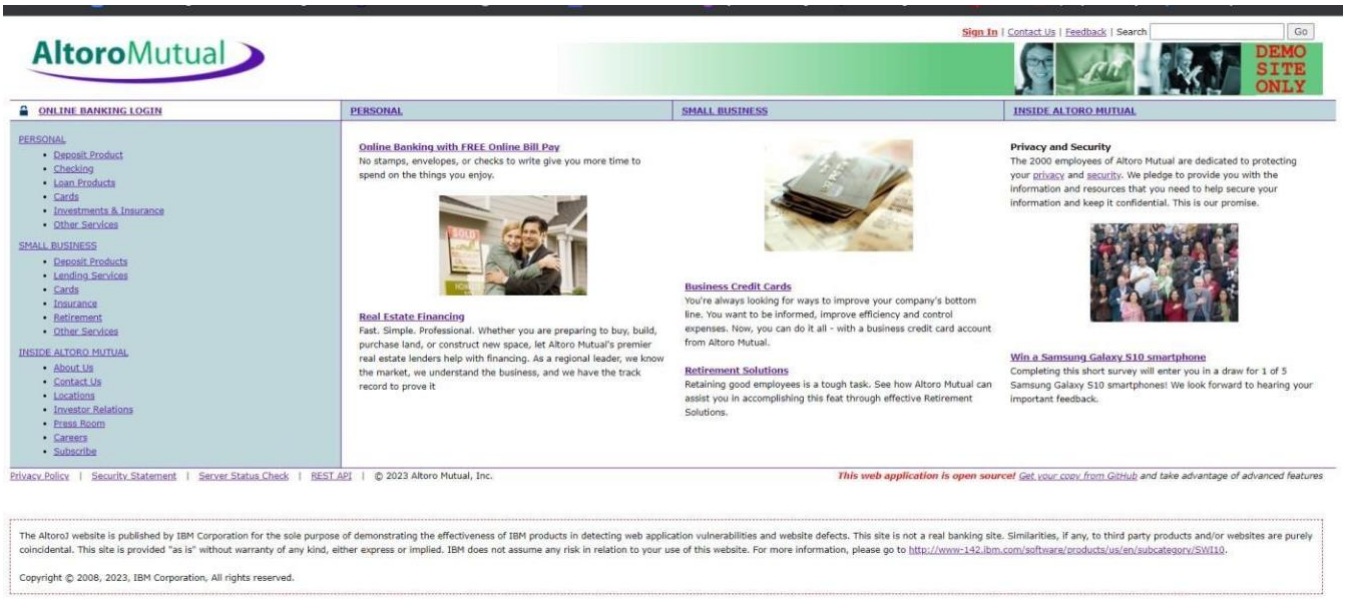
Business Impact: To effectively lessen the business consequences stemming from CWE-732, organizations must place a strong emphasis on bolstering their configuration management practices. This involves the establishment of secure default settings, the routine conduct of comprehensive security evaluations, and the deployment of intrusion detection systems to promptly detect and address misconfigurations and vulnerabilities. These actions are of paramount importance in safeguarding sensitive data, fortifying system security, and preserving the organization's esteemed reputation.

Vulnerability path: http://testfire.net/

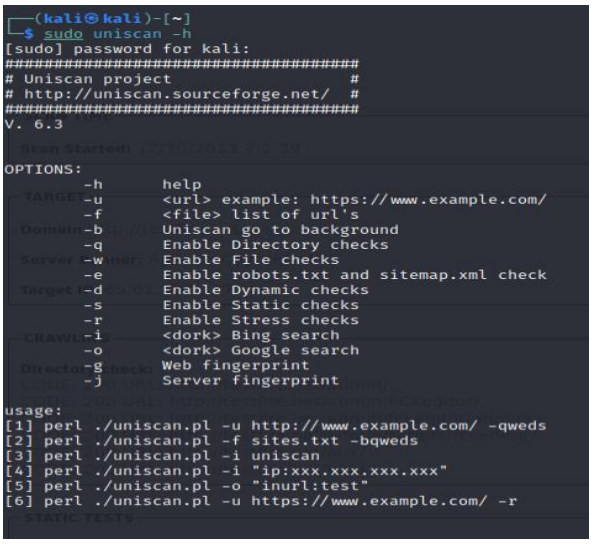
Vulnerability parameter: http://testfire.net/login.jsp

Steps to Reproduce:

- Access the URL



- We will be using the uniscan tool which is available in the kali.



```
┌──(kali㉿kali)-[~]
└─$ sudo uniscan -h
[sudo] password for kali:
#####################################
# Uniscan project                   #
# http://uniscan.sourceforge.net/   #
#####################################
V. 6.3

OPTIONS:
        -h      help
        -u      <url> example: https://www.example.com/
        -f      <file> list of url's
        -b      Uniscan go to background
        -q      Enable Directory checks
        -w      Enable File checks
        -e      Enable robots.txt and sitemap.xml check
        -d      Enable Dynamic checks
        -s      Enable Static checks
        -r      Enable Stress checks
        -i      <dork> Bing search
        -o      <dork> Google search
        -g      Web fingerprint
        -j      Server fingerprint

usage:
[1] perl ./uniscan.pl -u http://www.example.com/ -qweds
[2] perl ./uniscan.pl -f sites.txt -bqweds
[3] perl ./uniscan.pl -i uniscan
[4] perl ./uniscan.pl -i "ip:xxx.xxx.xxx.xxx"
[5] perl ./uniscan.pl -o "inurl:test"
[6] perl ./uniscan.pl -u https://www.example.com/ -r
```

- We will use this tool to scan for hidden directories that can be access due to poor security measures taken in this website.

```
┌──(kali㉿kali)-[~]
└─$ sudo uniscan -u http://testfire.net/ -qweds
[sudo] password for kali:
###################################
# Uniscan project                 #
# http://uniscan.sourceforge.net/ #
###################################
V. 6.3


Scan date: 17-10-2023 7:18:22
==============================================================================
| Domain: http://testfire.net/
| Server: Apache-Coyote/1.1
| IP: 65.61.137.117
==============================================================================
|
| Directory check:
| [+] CODE: 200 URL: http://testfire.net/admin/fckeditor/admins/
| [+] CODE: 200 URL: http://testfire.net/admin/
| [+] CODE: 200 URL: http://testfire.net/admin/scripts/fckeditor/
| [+] CODE: 200 URL: http://testfire.net/admin/FCKeditor/
| [+] CODE: 200 URL: http://testfire.net/aux/
| [+] CODE: 200 URL: http://testfire.net/bank/
==============================================================================
|
```

- We found directories that can be accessed easily. Also all the files that are present in the website has also been found.

```
|
| File check:
| [+] CODE: 200 URL: http://testfire.net/admin/account.asp
| [+] CODE: 200 URL: http://testfire.net/admin/account.html
| [+] CODE: 200 URL: http://testfire.net/admin/admin.php
| [+] CODE: 200 URL: http://testfire.net/admin/admin_phpinfo.php4
| [+] CODE: 200 URL: http://testfire.net/admin/account.php
| [+] CODE: 200 URL: http://testfire.net/admin/admin.shtml
| [+] CODE: 200 URL: http://testfire.net/admin/aindex.htm
| [+] CODE: 200 URL: http://testfire.net/admin/auth.php
| [+] CODE: 200 URL: http://testfire.net/admin/cfg/configscreen.inc.php+
| [+] CODE: 200 URL: http://testfire.net/admin/cfg/configsite.inc.php+
| [+] CODE: 200 URL: http://testfire.net/admin/cfg/configsql.inc.php+
| [+] CODE: 200 URL: http://testfire.net/admin/cfg/configtache.inc.php+
| [+] CODE: 200 URL: http://testfire.net/admin/cms/htmltags.php
| [+] CODE: 200 URL: http://testfire.net/admin/config.php
| [+] CODE: 200 URL: http://testfire.net/admin/contextAdmin/contextAdmin.html
| [+] CODE: 200 URL: http://testfire.net/admin/controlpanel.asp
| [+] CODE: 200 URL: http://testfire.net/admin/controlpanel.php
| [+] CODE: 200 URL: http://testfire.net/admin/controlpanel.html
| [+] CODE: 200 URL: http://testfire.net/admin/cplogfile.log
| [+] CODE: 200 URL: http://testfire.net/admin/cp.php
| [+] CODE: 200 URL: http://testfire.net/admin/cp.html
| [+] CODE: 200 URL: http://testfire.net/admin/database/wwForum.mdb
| [+] CODE: 200 URL: http://testfire.net/admin/credit_card_info.php
| [+] CODE: 200 URL: http://testfire.net/admin/datasource.asp
| [+] CODE: 200 URL: http://testfire.net/admin/db.php
| [+] CODE: 200 URL: http://testfire.net/admin/home.asp
| [+] CODE: 200 URL: http://testfire.net/admin/home.php
| [+] CODE: 200 URL: http://testfire.net/admin/index.asp
| [+] CODE: 200 URL: http://testfire.net/admin/index.html
| [+] CODE: 200 URL: http://testfire.net/admin/index.php
| [+] CODE: 200 URL: http://testfire.net/admin/phpinfo.php
| [+] CODE: 200 URL: http://testfire.net/admin/settings.inc.php+
| [+] CODE: 200 URL: http://testfire.net/admin/system_footer.php
| [+] CODE: 200 URL: http://testfire.net/admin/upload.php
| [+] CODE: 200 URL: http://testfire.net/admin/templates/header.php
| [+] CODE: 200 URL: http://testfire.net/admin/wg_user-info.ml
| [+] CODE: 200 URL: http://testfire.net/admin/script.php
| [+] CODE: 200 URL: http://testfire.net/login.jsp
==============================================================================
|
```

- Let us go through one webpage to see the vulnerability

- Here we can see the sensitive data of job applications which can help an attacker.

Recommendations:

- Implement a change management process to document and review all changes to system configurations. This helps prevent unauthorized or unintended alterations.
- Follow security guidelines provided by software and hardware vendors to ensure secure configurations.
- Regularly perform compliance checks against industry standards and regulations to ensure that configurations remain in compliance with security requirements.

8. Vulnerability Name: Cross-Site Request Forgery

CWS: 352

OWASP Category: A08: Cross-Site Request Forgery

Description:

Cross-Site Request Forgery (CSRF) is an attack that forces an end user to execute unwanted actions on a web application in which they're currently authenticated. With a little help of social engineering (such as sending a link via email or chat), an attacker may trick the users of a web application into executing actions of the attacker's choosing. If the victim is a normal user, a successful CSRF attack can force the user to perform state changing requests like transferring funds, changing their email address, and so forth. If the victim is an administrative account, CSRF can compromise the entire web application.

Business Impact:

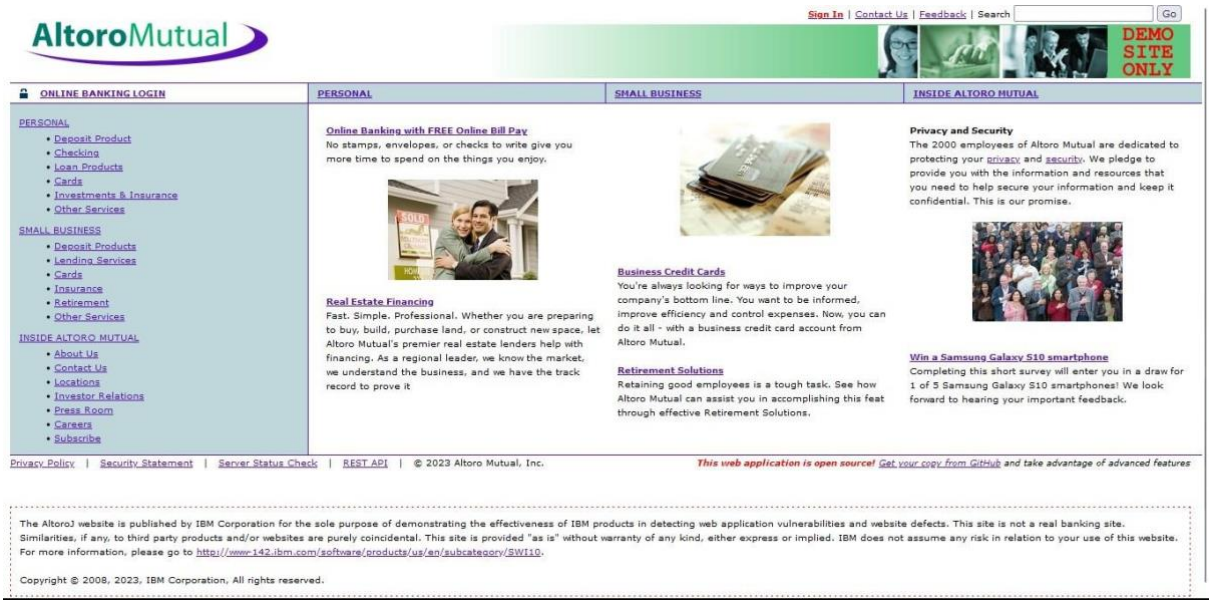
To effectively reduce the business repercussions of CWE-352, organizations must prioritize secure coding practices, including the incorporation of anti-CSRF tokens, while also conducting routine security assessments and testing to detect and rectify CSRF vulnerabilities. Furthermore, educating users on secure browsing practices is pivotal in preventing CSRF attacks. These actions are of paramount importance in the protection of data and in upholding the trust of customers and partners.

Vulnerability Path: http://testfire.net/

Vulnerability Parameter: http://testfire.net/bank/transfer.jsp

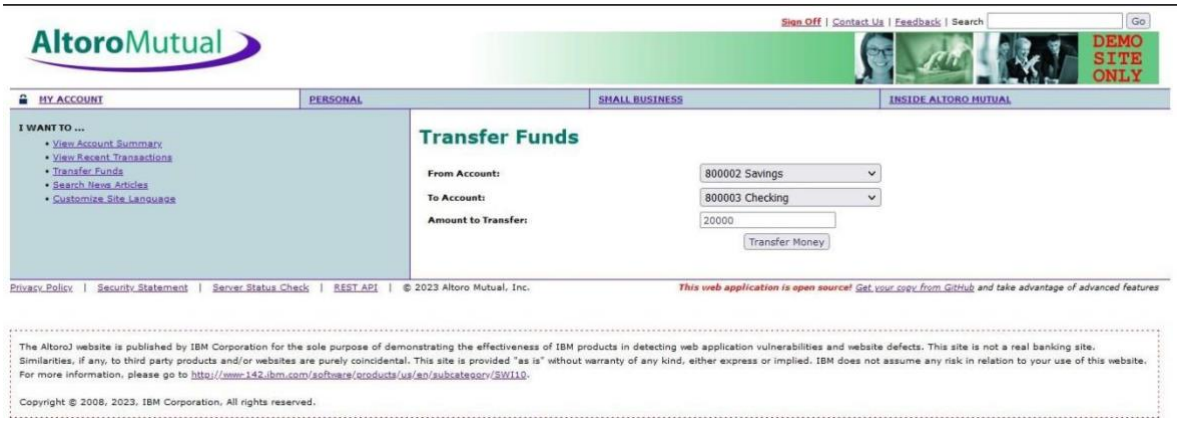
Steps to Reproduce:

- Open https://testfire.net/.



- Login to it using default credentials, and you will find two accounts.

- Capture the request in any Intercepting proxy (Burp Suite, Charles Proxy etc.) while transferring the amount from one account to another as shown in the below picture.



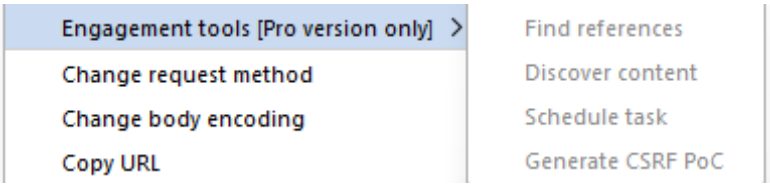
```
Pretty  Raw  Hex
1  POST /bank/doTransfer HTTP/1.1
2  Host: testfire.net
3  User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64; rv:109.0) Gecko/20100101 Firefox/119.0
4  Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,*/*;q=0.8
5  Accept-Language: en-US,en;q=0.5
6  Accept-Encoding: gzip, deflate, br
7  Content-Type: application/x-www-form-urlencoded
8  Content-Length: 80
9  Origin: http://testfire.net
10 Connection: close
11 Referer: http://testfire.net/bank/transfer.jsp
12 Cookie: JSESSIONID=14BE38CD21D31A5D4ED4DED118BGAF5E; AltoroAccounts=
   "ODAwMDAyflNhdmluZ3N+LTEuMTc5Mzg30TI5NTIwNzQ2RTQ0fDgwMDAwM35DaGVja2luZ34xLjE30TM4Nzky0TUyMDc0NkU0NHw0NTM5MDgyMDM5Mzk2Mjg4fkNy2WRpdCBDYXJkfiOyLjA50TU0MzQ
   wMTI30DgONjQ2RTE4fA==="
13 Upgrade-Insecure-Requests: 1
14
15 fromAccount=800002&toAccount=800003&transferAmount=20000&transfer=Transfer+Money
```

Inspector

- Request attributes 2
- Request query parameters 0
- Request body parameters 4
- Request cookies 2
- Request headers 12

Search 0 highlights

- Generate CSRF POC using Burp Suite Engagement tools (by modifying the account numbers accordingly)



- As the victim is already logged-in to the site an attacker can send the crafted malicious form/URL to the user to his email/text/by any other means and trick the victim using his social engineering skills/techniques like below.

```
Limited Time Offer Get a $100 Gift Card Now!!  (Evil :P)
```

- User thinks, he could use a free $100 gift card! What could go wrong? You open the email and click the link/open the attachment.

- Once the user opens the form a transaction is automatically performed from his account without his knowledge as the user is already logged in.

Recent Transactions

After [ ] Before [ ] Submit
yyyy-mm-dd yyyy-mm-dd

Transaction ID	Transaction Time	Account ID	Action	Amount
12539	2023-10-17 06:38	800002	Deposit	$20000.00
12538	2023-10-17 06:38	800003	Withdrawal	-$20000.00
12537	2023-10-17 06:36	800003	Deposit	$1234.00
12536	2023-10-17 06:36	800003	Withdrawal	-$1234.00
12535	2023-10-17 06:36	800003	Deposit	$1234.00
12534	2023-10-17 06:36	800003	Withdrawal	-$1234.00
12533	2023-10-17 06:36	800003	Deposit	$1234.00
12532	2023-10-17 06:36	800003	Withdrawal	-$1234.00

Recommendations:

- Implement Synchronizer Token Patterns to include unique, randomly generated tokens in each HTTP request and validate these tokens on the server side to verify the authenticity of the request.
- Utilize the SameSite attribute in cookies to restrict their usage to same-site requests only, preventing them from being sent along with cross-site requests and effectively mitigating CSRF attacks.

- Utilize framework-specific protections and security features to prevent CSRF attacks. Many modern web frameworks have built-in mechanisms and libraries for handling CSRF vulnerabilities effectively.
- Implement a Content Security Policy to restrict the sources from which various types of content can be loaded. By specifying the trusted sources of content, you can minimize the risk of malicious code execution and reduce the likelihood of successful CSRF attacks.

9. Vulnerability name: Cleartext Transmission of Sensitive Information

   CWE: 319

   Description: The product transmits sensitive or security-critical data in cleartext in a communication channel that can be sniffed by unauthorized actors. Many communication channels can be "sniffed" (monitored) by adversaries during data transmission. For example, in networking, packets can traverse many intermediary nodes from the source to the destination, whether across the internet, an internal network, the cloud, etc. Some actors might have privileged access to a network interface or any link along the channel, such as a router, but they might not be authorized to collect the underlying data. As a result, network traffic could be sniffed by adversaries, spilling security-critical data.

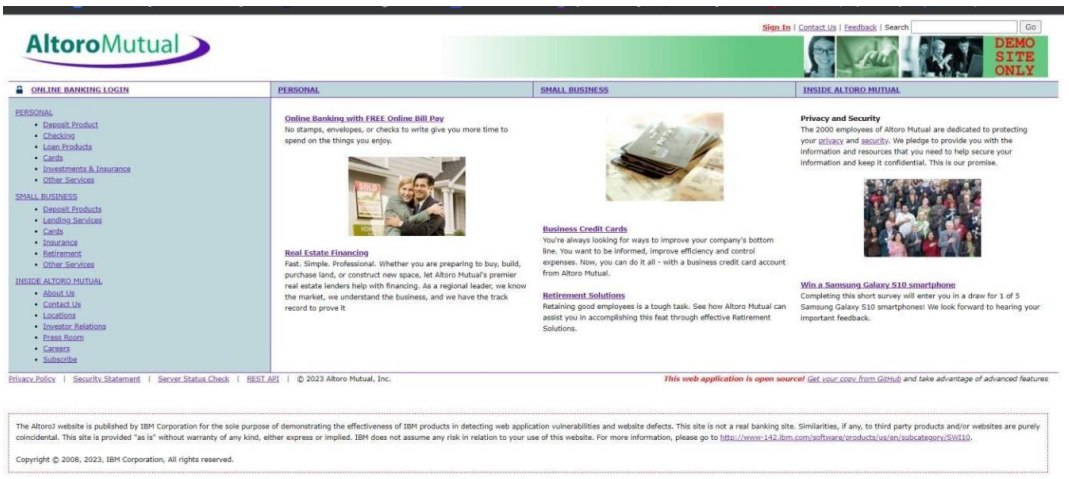
   Business Impact: To effectively reduce the business consequences associated with CWE-319, organizations must prioritize the adoption of secure data transmission practices. This includes the utilization of encryption and robust, secure protocols. The routine conduct of security assessments and testing is pivotal in pinpointing and remedying vulnerabilities linked to data transmission. Furthermore, the education of users on secure data handling practices plays a vital role in proactively preventing data exposure incidents. These actions are of paramount importance in the protection of sensitive data and in preserving the trust of both customers and partners.

   Vulnerability path: http://testfire.net/

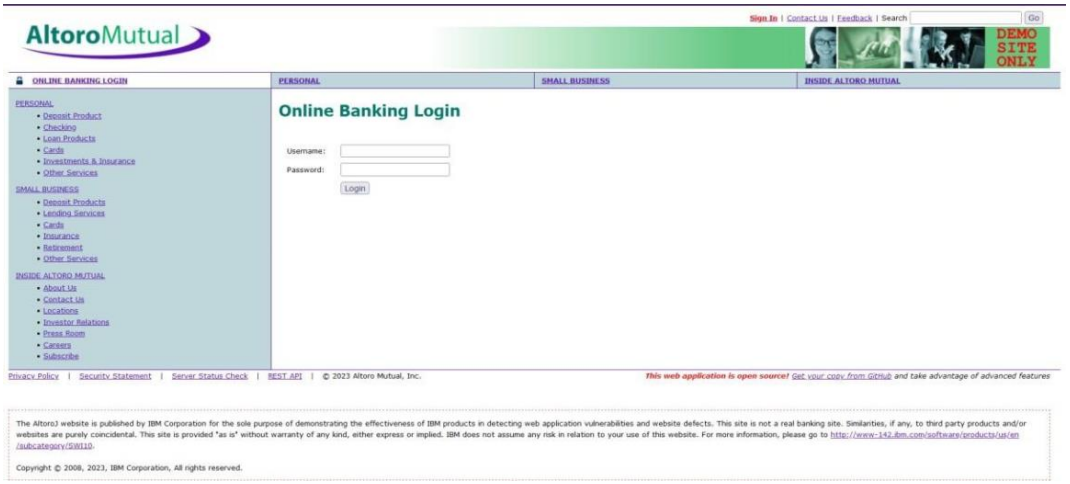
   Vulnerability parameter: http://testfire.net/login.jsp

   Steps to reproduce:

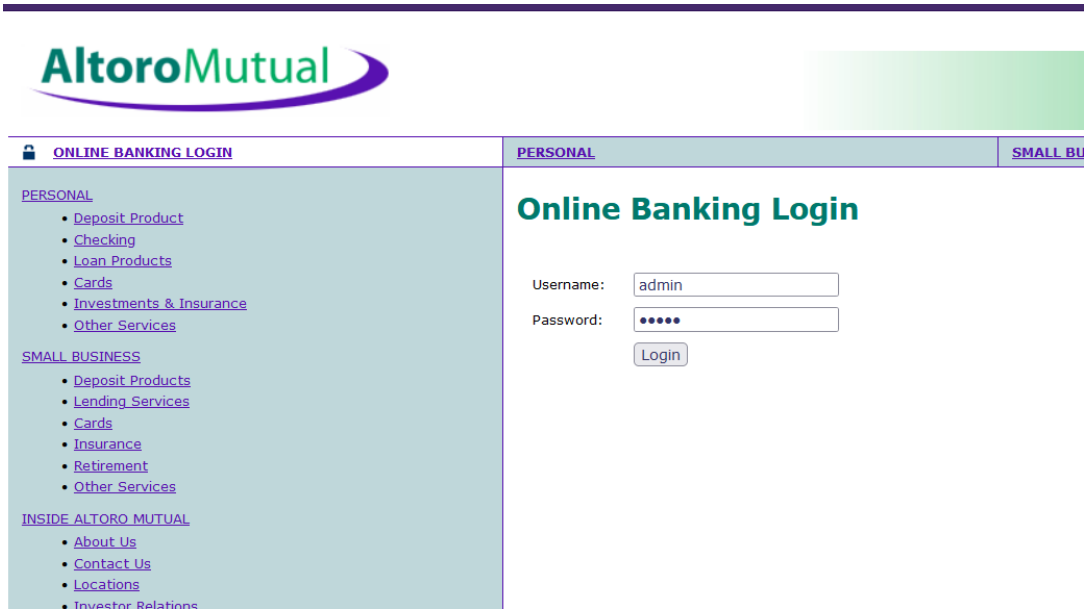
   - Access the URL



   - Now we will try to sign in to this website with admin privileges



- Also, we will be using burp suite to get requests from the website and know additional information.
- We use 'admin' for the username and password.



- This request has been received in the burp suite with the username and password as well in clear text.

```
Pretty   Raw   Hex
1  POST /doLogin HTTP/1.1
2  Host: testfire.net
3  User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64; rv:109.0) Gecko/20100101 Firefox/117.0
4  Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,*/*;q=0.8
5  Accept-Language: en-US,en;q=0.5
6  Accept-Encoding: gzip, deflate, br
7  Content-Type: application/x-www-form-urlencoded
8  Content-Length: 37
9  Origin: http://testfire.net
10 Connection: close
11 Referer: http://testfire.net/login.jsp
12 Cookie: JSESSIONID=B6ADB36ACD5C83083787343A1F97F853; AltoroAccounts="ODAwMDAwfkNvcnBvcmFOZX45LjQ30TA1MTE2MUU3fDgwMDAwMX5DaGVjaZluZ34tNC4yMjc0NjkONTZFN3w="
13 Upgrade-Insecure-Requests: 1
14
15 uid=admin&passw=admin&btnSubmit=Login
```

- Now we click on forward request in the burp suite and then we will be redirected to the admin user details. Here in the burp suite, we can clearly see the login details in clear text. this is the clear indication of the vulnerability which can lead to data breach, monitored, and manipulated as well.

Recommendations:

- Educate users and employees on secure data handling practices, such as recognizing secure websites (look for "https") and avoiding insecure Wi-Fi networks.
- Employ data masking or redaction techniques to replace sensitive data with placeholders or cryptographic representations during transmission.
- Implement encryption for data in transit. Use secure encryption protocols such as TLS/SSL to protect sensitive information during transmission.

- Use secure communication protocols for transmitting data, such as HTTPS for web applications, and ensure that the selected protocols are kept up-to-date.

10. Vulnerability name: Click Jacking

    CWE: 1021

    Description: The web application does not restrict or incorrectly restricts frame objects or UI layers that belong to another application or domain, which can lead to user confusion about which interface the user is interacting with.

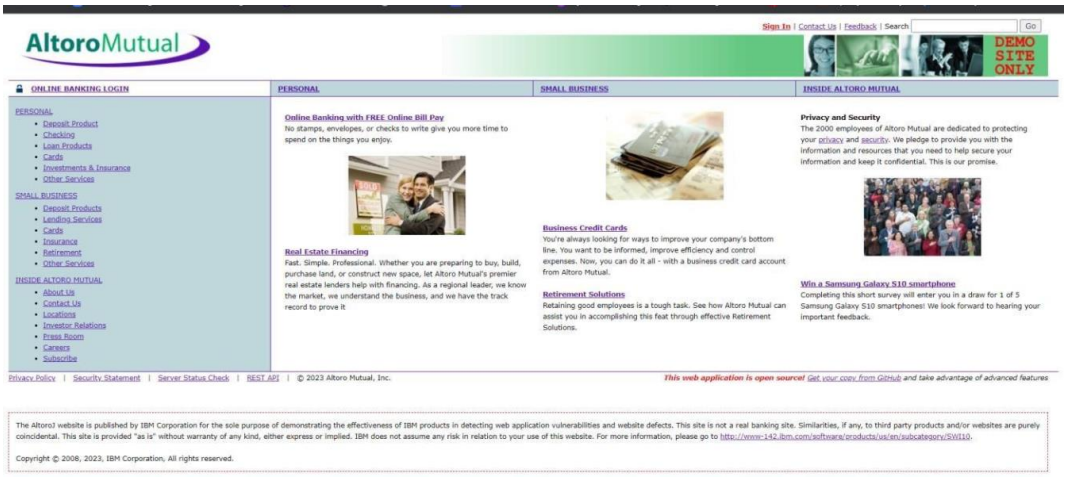
    Business Impact: To effectively reduce the business consequences resulting from CWE-1021, organizations must prioritize the implementation of protective measures, such as frame-busting code. Simultaneously, educating users on safe browsing practices plays a crucial role in preventing clickjacking incidents. Additionally, routine security assessments and testing are pivotal for identifying and mitigating vulnerabilities associated with clickjacking. These actions are of paramount importance in upholding user trust, ensuring data protection, and safeguarding the organization's reputation.

    Vulnerability path: http://testfire.net/

    Vulnerability parameter: http://testfire.net/

    Steps to reproduce:

    - Access the URL

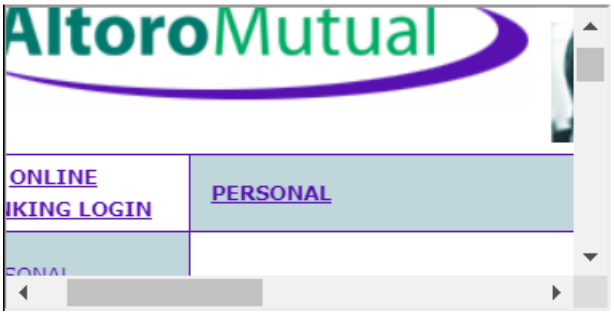


    - Then take the URL and use it for the html code
    - we will be writing some html code to perform this vulnerability. we will be writing the code in vs code for better flexibility and functionality.

```
<html>
<body>
<title>Click jacking vulnerability</title>
<h2>This website is vulnerable to clickjacking</h2>
<iframe  src="http://testfire.net/"></iframe>
</body>
</html>
```

    - After the code has been written then we will be executing the code in the browser.

This website is vulnerable to clickjacking



- From this image we can see that the vulnerability has been found.

Recommendation:

- Set the X-Frame-Options HTTP response header to deny or same-origin to control how your site can be framed. This is supported by most modern browsers.
- Utilize frame-busting JavaScript code in web applications to prevent the embedding of your site within malicious iframes. This code can disrupt clickjacking attempts.
- Implement a Content Security Policy to restrict which domains can embed your site in iframes. This can help prevent unauthorized framing.
- Implement additional security controls to prevent UI redress attacks, such as clickjacking, within your web application.

11. Vulnerability name: web server allows password auto-completion

CWE: 310

Description:
Weaknesses in this category are related to the design and implementation of data confidentiality and integrity. Frequently these deal with the use of encoding techniques, encryption libraries, and hashing algorithms. The weaknesses in this category could lead to a degradation of the quality data if they are not addressed.

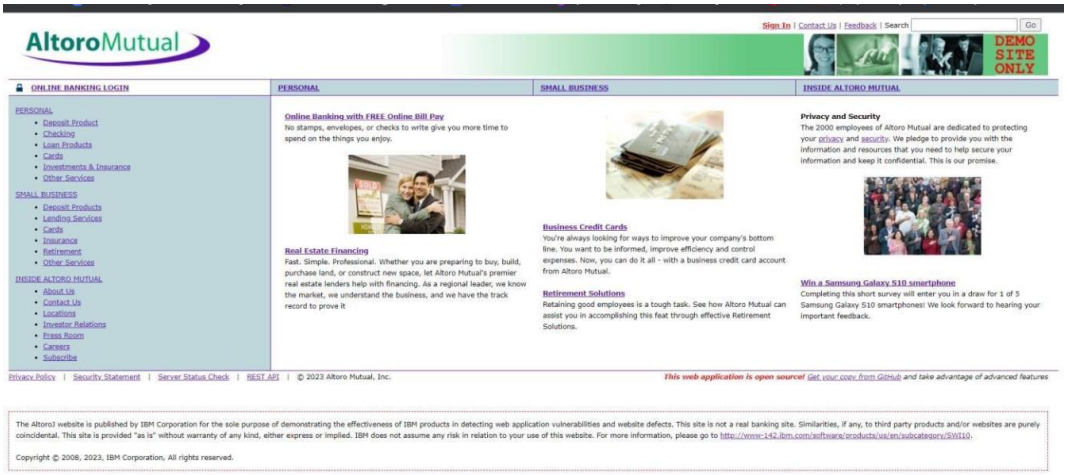
Business impact:
To effectively lessen the business consequences associated with CWE-310, organizations must prioritize the adoption of secure cryptographic practices. This entails ensuring proper password storage and robust encryption key management. Simultaneously, conducting routine security assessments and testing is pivotal in detecting and mitigating vulnerabilities related to cryptographic issues. Striving for compliance with pertinent data protection regulations and industry standards is equally essential. These actions are of paramount importance in the protection of sensitive data, the upholding of user trust, and the preservation of the organization's esteemed reputation.

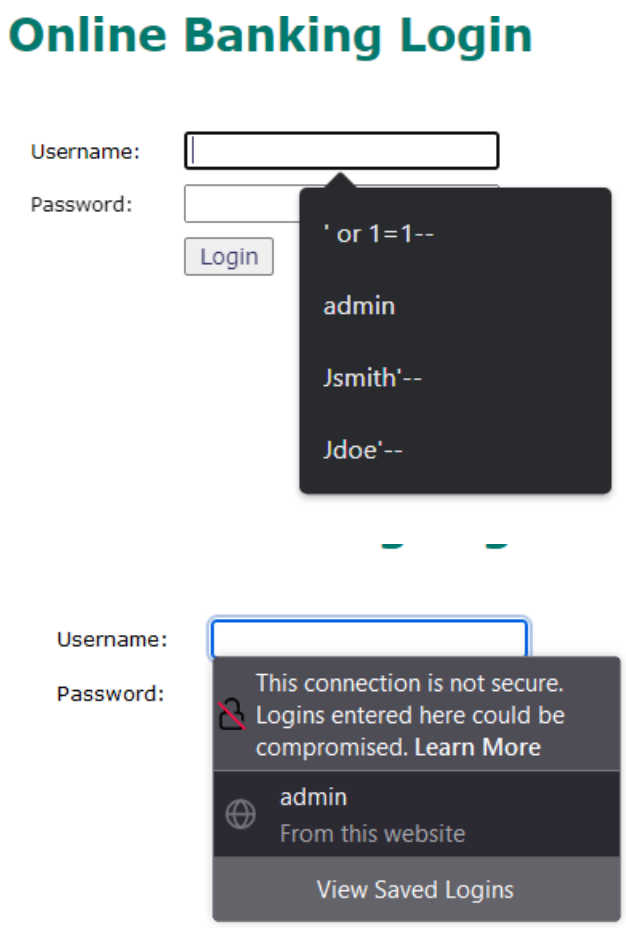
Vulnerability path: http://testfire.net/

Vulnerability parameter: http://testfire.net/login.jsp

Steps to reproduce:

- Access the URL





- From this image we can see the usernames and the passwords getting auto filled. This is a potential vulnerability as this can be a doorway for attackers.

Responsibilities:

- Implement secure password storage mechanisms, such as using strong and salted cryptographic hashing algorithms like bcrypt or scrypt.
- Implement robust encryption key management practices, including secure key storage, key rotation, and access controls.
- Utilize well-established and proven cryptographic libraries and algorithms for encryption and decryption.
- Use secure encryption protocols, like TLS/SSL, for data transmission over networks to protect data in transit.

12. Vulnerability name: Cross-site scripting (DOM based XSS)

CWE: 79

OWASP category: A03: Injections

Description: It occurs when a malicious script is injected directly into a vulnerable web application. Reflected XSS involves the reflection of a malicious script of a web application, onto a user's browser.

DOM-XSS: DOM stands for Document Object Model and is a programming interface for HTML and XML documents. It represents that programs can change the document structure, style and content. A webpage is a document and this document can be either displayed in the browser window or as the HTML Source.

Business Impact: The application stores dangerous data in a database, message forum, visitor log, or other trusted data store. Later, the dangerous data is subsequently read back into the application and included in dynamic content. From an attacker's perspective, the optimal place to inject malicious content is in an area displayed to many users or particularly interesting users. Interesting users typically have elevated privileges in the application or interact with sensitive data that is valuable to the attacker. If one of these users executes malicious content, the attacker may be able to perform privileged operations on behalf of the user or gain access to sensitive data belonging to the user.

Vulnerability path: https://vit.ac.in/

Vulnerability parameter: https://vit.ac.in/

Steps to Reproduce:

- Access the URL

- We will try to perform the DOM XSS attack which is displaying the source code of the webpage.

- Now we can access the source code of the webpage in two ways by clicking on the inspect option or by simply pressing Ctrl + U.

```
Line wrap
<!DOCTYPE html>

<html lang="en">
<head>
<meta charset="utf-8">
<meta http-equiv="X-UA-Compatible" content="IE=edge,chrome=1">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
  <meta http-equiv="Content-Type" content="text/html; charset=utf-8" />
<link rel="alternate" type="application/rss+xml" title="VIT RSS" href="https://vit.ac.in/rss.xml" />
<link rel="shortcut icon" href="https://vit.ac.in/sites/all/themes/vittheme/favicon.ico" type="image/vnd.microsoft.icon" />
<meta name="description" content="One of the Best Educational Institution in India. Maintained an excellent placement records and gives training students to get
<meta name="abstract" content="VIT, the top university in India was established with the aim of providing quality higher education on par with international stan
<meta name="keywords" content="vit, vellore institute of technology, viteee, vit university, vit vellore, top engineering colleges in india, top 10 engineering c
<meta name="generator" content="Drupal" />
<link rel="canonical" href="https://vit.ac.in/" />
<meta property="og:site_name" content="Vellore Institute of Technology" />
<meta property="og:type" content="website" />
<meta property="og:url" content="https://vit.ac.in/" />
<meta property="og:title" content="Vellore Institute of Technology | A Place to Learn, Chance to grow" />
<meta property="og:description" content="One of the Best Educational Institution in India. Maintained an excellent placement records and gives training students
<meta property="og:image" content="https://vit.ac.in/sites/all/themes/vittheme/images/og-vit-updated.jpg" />
<meta property="og:image:url" content="https://vit.ac.in/sites/all/themes/vittheme/images/og-vit-updated.jpg" />
<meta name="dcterms.title" content="VIT" />
<meta name="dcterms.type" content="Text" />
<meta name="dcterms.format" content="text/html" />
<meta name="dcterms.identifier" content="https://vit.ac.in/" />
<meta name="facebook-domain-verification" content="9bbekg8iy8yy964rj4euqpwennaxoc" />
      <title>Vellore Institute of Technology | A Place to Learn, Chance to grow</title>
  <style type="text/css" media="screen">
@import url("https://vit.ac.in/sites/all/themes/vittheme/css/bootstrap.min.css?rdiap4");
@import url("https://vit.ac.in/sites/all/themes/vittheme/css/style.css?rdiap4");
@import url("https://vit.ac.in/sites/all/themes/vittheme/css/theme.css?rdiap4");
@import url("https://vit.ac.in/sites/all/themes/vittheme/css/combine.css?rdiap4");
@import url("https://vit.ac.in/sites/all/themes/vittheme/css/responsive.css?rdiap4");
</style>
<meta name="author" content="VIT University">
<meta http-equiv="Content-Type" content="text/html; charset=utf-8">
<meta name="language" content="English">
<meta name="copyright" content="https://vit.ac.in/"/>
<meta name="geo.region" content="IN-TN" />
<meta name="geo.placename" content="Vellore" />
```

- Many programs can help to alter this source code to find vulnerable spots.

- So, getting access to the source code of the webpage can lead to potential vulnerabilities as attackers can abuse the loop holes, exploits, vulnerabilities in the code and act accordingly.

Recommendation:

- When interacting with databases, use parameterized statements or prepared statements to avoid SQL injection, which can be a vector for XSS.
- Sanitize and validate all user inputs and ensure that any data displayed on the web page is properly encoded. Use output encoding libraries or functions to prevent script injection.
- Avoid using DOM manipulation methods that can introduce vulnerabilities. Be cautious with inner HTML, document. Write, and other methods that can execute scripts.
- Consider using well-established JavaScript frameworks and libraries that include built-in security features to prevent DOM XSS.

13. Vulnerability name: JQuery 1.2 < 3.5.0 Multiple XSS

CWE: 79

OWASP category: A03: Injections

Description: It occurs when a malicious script is injected directly into a vulnerable web application. Reflected XSS involves the reflection of a malicious script of a web application, onto a user's browser.

Stored XSS (Persistent XSS): In this type of attack, the malicious script is permanently stored on a server or in a database, often in user-generated content like comments or forum posts. When other users view the affected content, the script executes in their browsers, potentially compromising their data or sessions.

Reflected XSS: Reflected XSS occurs when the malicious script is embedded in a URL and immediately executed when a victim clicks on a manipulated link. The script is not stored on a server but is reflected off a web application, making it a one-time attack.

DOM-based XSS: DOM stands for Document Object Model and is a programming interface for HTML and XML documents. It represents that programs can change the document structure, style and content. A webpage is a document and this document can be either displayed in the browser window or as the HTML Source.

Business Impact: The application stores dangerous data in a database, message forum, visitor log, or other trusted data store. Later, the dangerous data is subsequently read back into the application and included in dynamic content. From an attacker's perspective, the optimal place to inject malicious content is in an area displayed to many users or particularly interesting users. Interesting users typically have elevated privileges in the application or interact with sensitive data that is valuable to the attacker. If one of these users executes malicious content, the attacker may be able to perform privileged operations on behalf of the user or gain access to sensitive data belonging to the user.

Vulnerability path: https://vit.ac.in/

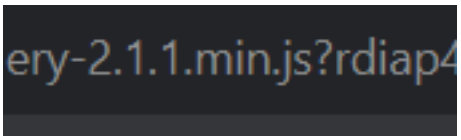
Vulnerability parameter: https://vit.ac.in/sites/all/themes/vittheme/js/jquery-2.1.1.min.js?rdiap4

Steps to Reproduce:

- Access the URL

- Try to access the URL https://vit.ac.in/sites/all/themes/vittheme/js/jquery- . This will lead us to the webpage with jquery script in it.

- We can also see the jquery version that has been used in the website



- We can access the source code of the jquery script that has been used in making the website, which can lead to multiple potential threats

Recommendation:

- The most important recommendation is to update to the latest version of jQuery. Newer versions typically include security patches, bug fixes, and performance improvements.
- Stay vigilant about jQuery updates and security announcements. Subscribe to relevant security mailing lists or forums to receive notifications about new releases.
- A Content Security Policy (CSP) can help mitigate various types of client-side vulnerabilities, including Cross-Site Scripting (XSS). Consider implementing a CSP in your web application to restrict the sources from which scripts can be loaded.
- If updating jQuery is not immediately possible, review your code for potential security issues. Ensure that you are escaping or sanitizing user inputs and that you are using safe coding practices to prevent XSS and other security vulnerabilities.

14. Vulnerability name: Information Exposure (Web Server HTTP Header Internal IP Disclosure)

CWE: 200

Description: The product exposes sensitive information to an actor that is not explicitly authorized to have access to that information.

Business Impact: To effectively reduce the business consequences associated with CWE-200, organizations must prioritize the implementation of robust data access controls, secure data handling practices, and privacy protection measures. Regular security assessments and testing are instrumental in detecting and rectifying vulnerabilities related to information exposure. Furthermore, strict adherence to data protection regulations and industry standards is indispensable. These actions are of paramount importance in the protection of sensitive data, the preservation of user trust, and the upholding of the organization's esteemed reputation.

Vulnerability path: https://vit.ac.in/

Vulnerability parameter: https://vit.ac.in/

Steps to Reproduce:

- Access the URL

- We will be using a tool to perform a scan on this website. (Nessus Scanner)

LOW **Web Server HTTP Header Internal IP Disclosure**

Description

This may expose internal IP addresses that are usually hidden or masked behind a Network Address Translation (NAT) Firewall or proxy server.

There is a known issue with Microsoft IIS 4.0 doing this in its default configuration. This may also affect other web servers, web applications, web proxies, load balancers and through a variety of misconfigurations related to redirection.

Solution

Apply configuration suggested by vendor.

- Through this tool we can see the vulnerability in the scan along with detailed information regarding that vulnerability.

```
Nessus was able to exploit the issue using the following request :

GET / HTTP/1.0
Accept-Charset: iso-8859-1,utf-8;q=0.9,*;q=0.1
Accept-Language: en
Connection: Close
User-Agent: Mozilla/4.0 (compatible; MSIE 8.0; Windows NT 5.1; Trident/4.0)
Pragma: no-cache
Accept: image/gif, image/x-xbitmap, image/jpeg, image/pjpeg, image/png, */*



This produced the following truncated output (limited to 10 lines) :
------------------------------ snip ------------------------------
Location: https://10.10.7.35/
Content-Length: 0


------------------------------ snip ------------------------------
less...
```

- Now this image tells us the internal IP address of the target website, the attackers can easily exploit the website using the ip address.

Recommendation:

- Carefully review your web server's configuration files (e.g., Apache's httpd.conf, Nginx's nginx.conf) to identify anywhere internal IP addresses are inadvertently included in HTTP response headers.
- Remove or replace any internal IP addresses in the server configuration files with appropriate placeholders or public IP addresses.
- If you use proxy servers or load balancers, ensure that they are configured to hide internal IP addresses and only expose public IP addresses instances in HTTP headers.
- Review and configure HTTP response headers (e.g., Server, X-Powered-By) to limit the exposure of server and technology details, reducing the potential for disclosing internal IP addresses.

15. Vulnerability name: Information Exposure (SSL (Multiple Issues))

CWE: 200

Description: The product exposes sensitive information to an actor that is not explicitly authorized to have access to that information.

Business Impact: To effectively reduce the business consequences arising from SSL issues, organizations must give utmost priority to establishing a secure SSL/TLS configuration, conducting frequent security assessments and testing. Adhering to data protection regulations and industry standards is imperative. These actions play a pivotal role in safeguarding sensitive data, upholding user trust, and safeguarding the organization's esteemed reputation.

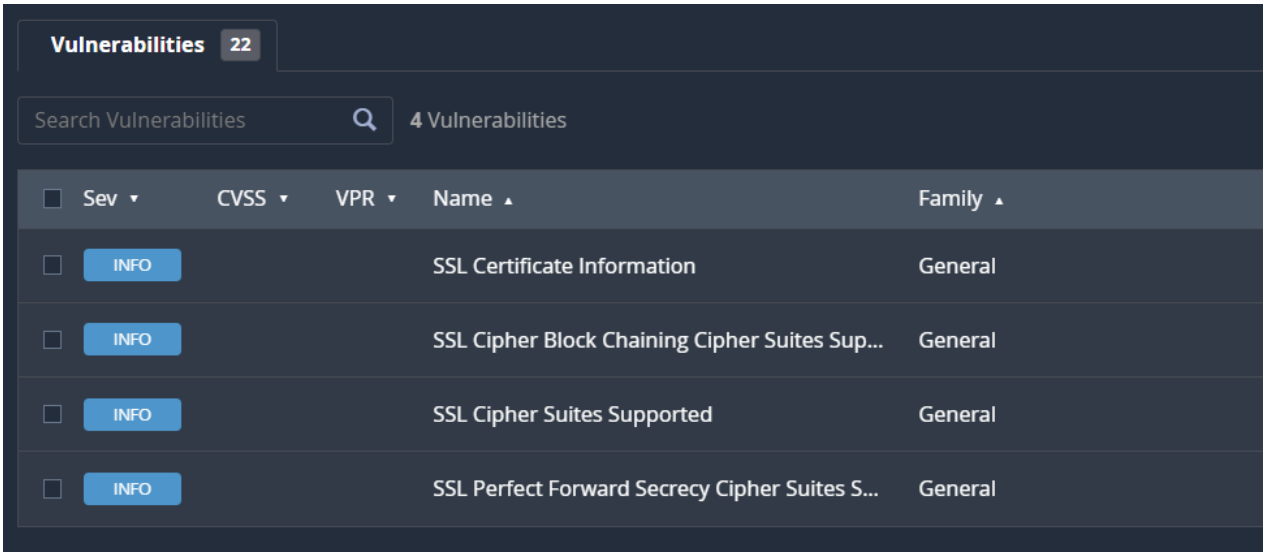
Vulnerability path: https://vit.ac.in/

Vulnerability parameter: https://vit.ac.in/

Steps to Reproduce:

- Access the URL

- We will be using a tool to perform a scan on this website. (Nessus Scanner)



- Through this tool we can see the vulnerability in the scan along with detailed information regarding that vulnerability.

INFO SSL Certificate Information

Description

This plugin connects to every SSL-related port and attempts to extract and dump the X.509 certificate.

Output

```
Subject Name:

Common Name: *.vit.ac.in

Issuer Name:

Country: GB
State/Province: Greater Manchester
Locality: Salford
Organization: Sectigo Limited
Common Name: Sectigo RSA Domain Validation Secure Server CA

Serial Number: 46 53 67 B6 23 C5 BE EE B9 6E E2 C0 5F 46 C9 F7

Version: 3

Signature Algorithm: SHA-256 With RSA Encryption

Not Valid Before: Sep 04 00:00:00 2023 GMT
Not Valid After: Aug 03 23:59:59 2024 GMT

Public Key Info:

Algorithm: RSA Encryption
Key Length: 2048 bits
```

- Now this image tells us the type of encryption used in the target website with the public key details as well.

INFO SSL Cipher Block Chaining Cipher Suites Supported

Description

The remote host supports the use of SSL ciphers that operate in Cipher Block Chaining (CBC) mode. These cipher suites offer additional security over Electronic Codebook (ECB) mode, but have the potential to leak information if used improperly.

INFO SSL Cipher Suites Supported

Description

This plugin detects which SSL ciphers are supported by the remote service for encrypting communications.

INFO SSL Perfect Forward Secrecy Cipher Suites Supported

Description

The remote host supports the use of SSL ciphers that offer Perfect Forward Secrecy (PFS) encryption. These cipher suites ensure that recorded SSL traffic cannot be broken at a future date if the server's private key is compromised.

- Through this we can understand what different types of encryptions are used in the website.

```
Here is the list of SSL CBC ciphers supported by the remote server :

  High Strength Ciphers (>= 112-bit key)

    Name                          Code            KEX          Auth     Encryption             MAC
    ----------------------        ----------      ---          ----     --------------------   ---
    ECDHE-RSA-CAMELLIA-CBC-128    0xC0, 0x76      ECDH         RSA      Camellia-CBC(128)      SHA256
    ECDHE-RSA-CAMELLIA-CBC-256    0xC0, 0x77      ECDH         RSA      Camellia-CBC(256)      SHA384
    DHE-RSA-AES128-SHA            0x00, 0x33      DH           RSA      AES-CBC(128)           SHA1
    DHE-RSA-AES256-SHA            0x00, 0x39      DH           RSA      AES-CBC(256)           SHA1
    DHE-RSA-CAMELLIA128-SHA       0x00, 0x45      DH           RSA      Camellia-CBC(128)      SHA1
    DHE-RSA-CAMELLIA256-SHA       0x00, 0x88      DH           RSA      Camellia-CBC(256)      SHA1
    DHE-RSA-SEED-SHA              0x00, 0x9A      DH           RSA      SEED-CBC(128)          SHA1
    ECDHE-RSA-AES128-SHA          0xC0, 0x13      ECDH         RSA      AES-CBC(128)           SHA1
    ECDHE-RSA-AES256-SHA          0xC0, 0x14      ECDH         RSA      AES-CBC(256)           SHA1
    AES128-SHA                    0x00, 0x2F      RSA          RSA      AES-CBC(128)           SHA1
    AES256-SHA                    0x00, 0x35      RSA          RSA      AES-CBC(256)           SHA1
    CAMELLIA128-SHA               0x00, 0x41      RSA          RSA      Camellia-CBC(128)      SHA1
    CAMELLIA256-SHA               0x00, 0x84      RSA          RSA      Camellia-CBC(256)      SHA1
    SEED-SHA                      0x00, 0x96      RSA          RSA      SEED-CBC(128)          SHA1
    DHE-RSA-AES128-SHA256         0x00, 0x67      DH           RSA      AES-CBC(128)           SHA256
    DHE-RSA-AES256-SHA256         0x00, 0x6B      DH           RSA      AES-CBC(256)           SHA256
    DHE-RSA-CAMELLIA128-SHA256    0x00, 0xBE      DH           RSA      Camellia-CBC(128)      SHA256
    DHE-RSA-CAMELLIA256-SHA256    0x00, 0xC4      DH           RSA      Camellia-CBC(256)      SHA256
    ECDHE-RSA-AES128-SHA256       0xC0, 0x27      ECDH         RSA      AES-CBC(128)           SHA256
    ECDHE-RSA-AES256-SHA384       0xC0, 0x28      ECDH         RSA      AES-CBC(256)           SHA384
    RSA-AES128-SHA256             0x00, 0x3C      RSA          RSA      AES-CBC(128)           SHA256
    RSA-AES256-SHA256             0x00, 0x3D      RSA          RSA      AES-CBC(256)           SHA256
    RSA-CAMELLIA128-SHA256        0x00, 0xBA      RSA          RSA      Camellia-CBC(128)      SHA256
    RSA-CAMELLIA256-SHA256        0x00, 0xC0      RSA          RSA      Camellia-CBC(256)      SHA256
```

Recommendation:

- Configure your web server to use the latest and strongest encryption protocols, such as TLS 1.2 or higher. Disable outdated and insecure protocols like SSLv3.
- Implement Perfect Forward Secrecy to ensure that even if the private key is compromised, previously recorded encrypted traffic remains secure.
- Safeguard your private keys by storing them securely and ensuring they are not accessible to unauthorized parties.

16. Vulnerability name: Information Exposure (HTTP (Multiple Issues))

CWE: 200

Description: The product exposes sensitive information to an actor that is not explicitly authorized to have access to that information.

Business Impact: To effectively reduce the business repercussions of HTTP issues, organizations must place a strong emphasis on ongoing monitoring and enhancement of their web services. Consistent performance testing and security assessments, coupled with proactive maintenance, are indispensable to ensure a seamless and secure online presence. Furthermore, offering timely and helpful customer support and transparent communication when HTTP issues arise is instrumental in mitigating adverse effects on user trust and satisfaction.

Vulnerability path: https://vit.ac.in/

Vulnerability parameter: https://vit.ac.in/

Steps to Reproduce:

- Access the URL

- We will be using a tool to perform a scan on this website. (Nessus Scanner)

Search Vulnerabilities | 2 Vulnerabilities

Sev	CVSS	VPR	Name	Family
INFO			HyperText Transfer Protocol (HTTP) Informa...	Web Servers
INFO			HTTP Server Type and Version	Web Servers

INFO HyperText Transfer Protocol (HTTP) Information

Description

This test gives some information about the remote HTTP protocol - the version used, whether HTTP Keep-Alive and HTTP pipelining are enabled, etc...

This test is informational only and does not denote any security problem.

```
Response Code : HTTP/1.1 307 Moved Temporarily

Protocol version : HTTP/1.1
SSL : no
Keep-Alive : no
Options allowed : (Not implemented)
Headers :

  Location: https://vit.ac.in/
  Content-Length: 0

Response Body :
```

- From the above image we can see the details of protocol version of the website and its status.

INFO HTTP Server Type and Version

Description

This plugin attempts to determine the type and the version of the remote web server.

```
The remote web server type is :

Apache
```

- We can see the type of remote webserver here.
- All this information can lead to multiple potential threats such as data breaches by hackers.

Recommendation:

- Minimize the use of unnecessary redirects as they can increase page load times. Use 301 (permanent) redirects for consistent content.
- Regularly check for broken links (404 errors) and fix them to ensure users find the content they are looking for.
- Properly configure CORS to control which domains are allowed to make requests to your server.
- Migrate your website from HTTP to HTTPS to ensure secure data transmission. This is essential for user trust and search engine ranking.

17. Vulnerability name: OS Identification Failed

CWE: 200

Description: OS identification failed refers to the inability of a system or software to accurately recognize the operating system running on a device or network. This issue can result from various factors such as misconfigurations, network issues, or compatibility problems. The failure to identify the OS correctly can lead to security vulnerabilities, hindered software updates, and potential operational disruptions. Resolving this issue promptly is crucial to ensure accurate security measures and the smooth functioning of the system or network.

Business Impact: It may lead to operational inefficiencies, potential security vulnerabilities, non-compliance with industry regulations, and a possible loss of customer trust. Such failures could result in downtime, compromised data security, regulatory penalties, and damage to the company's reputation, potentially leading to financial losses and decreased market competitiveness. Resolving this issue promptly is crucial to mitigate these risks and maintain the smooth functioning of the organization.

Vulnerability path: https://vit.ac.in/

Vulnerability parameter: https://vit.ac.in/

Steps to Reproduce:

- Access the URL

- We will be using a tool to perform a scan on this website. (Nessus Scanner)

INFO **OS Identification Failed**

Description

Using a combination of remote probes (TCP/IP, SMB, HTTP, NTP, SNMP, etc), it was possible to gather one or more fingerprints from the remote system. Unfortunately, though, Nessus does not currently know how to use them to identify the overall system.

```
If you think these signatures would help us improve OS fingerprinting,
please send them to :

  os-signatures@nessus.org

Be sure to include a brief description of the device itself, such as
the actual operating system or product / model names.


SSLcert:!:i/CN:Sectigo RSA Domain Validation Secure Server CAi/O:Sectigo
Limiteds/CN:*.vit.ac.in
02ed800c9e7f764615d7e9ffd74b643a6cace446

SinFP:!:
   P1:B10113:F0x12:W65535:O0204ffff:M496:
   P2:B10113:F0x12:W65535:O0204ffff010303030101040201010 80affffffff44454144:M1300:
   P3:B00000:F0x00:W0:O0:M0
   P4:190701_7_p=80R
```

- From the above image, we can see the details of SSL Certificate Information and SinFP Information.
- If it is released without proper authorization, it can lead to serious consequences such as increased vulnerability to cyberattacks, potential data breaches, privacy violations, and legal repercussions.
- All this information can lead to multiple potential threats such as data breaches by hackers.

Recommendations:

- Employ robust data encryption methods for both data in transit and at rest.
- Implement strict access controls and authentication protocols to limit unauthorized access.
- Use secure communication channels, such as encrypted email and file transfer protocols.
- Provide comprehensive training to employees on data security best practices.
- Conduct regular security audits and assessments to identify and address vulnerabilities promptly.
- Ensure compliance with relevant data protection regulations and have a well-defined incident response plan in place.

NESSUS SCAN REPORT:

testfire.net

0	0	3	1	24
CRITICAL	HIGH	MEDIUM	LOW	INFO

Vulnerabilities — Total: 28

SEVERITY	CVSS V3.0	VPR SCORE	PLUGIN	NAME
MEDIUM	6.5	-	142960	HSTS Missing From HTTPS Server (RFC 6797)
MEDIUM	6.5	-	104743	TLS Version 1.0 Protocol Detection
MEDIUM	6.5	-	157288	TLS Version 1.1 Protocol Deprecated
LOW	3.7	4.5	83875	SSL/TLS Diffie-Hellman Modulus <= 1024 Bits (Logjam)
INFO	N/A	-	46180	Additional DNS Hostnames
INFO	N/A	-	39446	Apache Tomcat Detection
INFO	N/A	-	45590	Common Platform Enumeration (CPE)
INFO	N/A	-	54615	Device Type
INFO	N/A	-	84502	HSTS Missing From HTTPS Server
INFO	N/A	-	10107	HTTP Server Type and Version
INFO	N/A	-	24260	HyperText Transfer Protocol (HTTP) Information
INFO	N/A	-	11219	Nessus SYN scanner
INFO	N/A	-	19506	Nessus Scan Information
INFO	N/A	-	11936	OS Identification
INFO	N/A	-	10919	Open Port Re-check
INFO	N/A	-	56984	SSL / TLS Versions Supported
INFO	N/A	-	10863	SSL Certificate Information
INFO	N/A	-	95631	SSL Certificate Signed Using Weak Hashing Algorithm (Known CA)
INFO	N/A	-	70544	SSL Cipher Block Chaining Cipher Suites Supported

INFO	N/A	-	21643	SSL Cipher Suites Supported
INFO	N/A	-	57041	SSL Perfect Forward Secrecy Cipher Suites Supported
INFO	N/A	-	94761	SSL Root Certification Authority Certificate Information
INFO	N/A	-	156899	SSL/TLS Recommended Cipher Suites
INFO	N/A	-	22964	Service Detection
INFO	N/A	-	25220	TCP/IP Timestamps Supported
INFO	N/A	-	121010	TLS Version 1.1 Protocol Detection
INFO	N/A	-	136318	TLS Version 1.2 Protocol Detection
INFO	N/A	-	10287	Traceroute Information

* indicates the v3.0 score was not available; the v2.0 score is shown

vit.ac.in

CRITICAL	HIGH	MEDIUM	LOW	INFO
0	0	1	1	26

Vulnerabilities — Total: 28

SEVERITY	CVSS V3.0	VPR SCORE	PLUGIN	NAME
MEDIUM	6.1	5.7	136929	JQuery 1.2 < 3.5.0 Multiple XSS
LOW	3.1	2.2	10759	Web Server HTTP Header Internal IP Disclosure
INFO	N/A	-	48204	Apache HTTP Server Version
INFO	N/A	-	45590	Common Platform Enumeration (CPE)
INFO	N/A	-	10107	HTTP Server Type and Version
INFO	N/A	-	24260	HyperText Transfer Protocol (HTTP) Information
INFO	N/A	-	106658	JQuery Detection
INFO	N/A	-	11219	Nessus SYN scanner
INFO	N/A	-	19506	Nessus Scan Information
INFO	N/A	-	42823	Non-compliant Strict Transport Security (STS)
INFO	N/A	-	50350	OS Identification Failed
INFO	N/A	-	66334	Patch Report
INFO	N/A	-	56984	SSL / TLS Versions Supported
INFO	N/A	-	10863	SSL Certificate Information
INFO	N/A	-	70544	SSL Cipher Block Chaining Cipher Suites Supported
INFO	N/A	-	21643	SSL Cipher Suites Supported
INFO	N/A	-	57041	SSL Perfect Forward Secrecy Cipher Suites Supported
INFO	N/A	-	94761	SSL Root Certification Authority Certificate Information
INFO	N/A	-	156899	SSL/TLS Recommended Cipher Suites

INFO	N/A	-	22964	Service Detection
INFO	N/A	-	42822	Strict Transport Security (STS) Detection
INFO	N/A	-	25220	TCP/IP Timestamps Supported
INFO	N/A	-	84821	TLS ALPN Supported Protocol Enumeration
INFO	N/A	-	136318	TLS Version 1.2 Protocol Detection
INFO	N/A	-	10287	Traceroute Information
INFO	N/A	-	100669	Web Application Cookies Are Expired
INFO	N/A	-	10386	Web Server No 404 Error Code Check
INFO	N/A	-	10302	Web Server robots.txt Information Disclosure

* indicates the v3.0 score was not available; the v2.0 score is shown

"With the Nessus scan report now at our disposal, we're equipped with critical data on the existing vulnerabilities and potential threats. This invaluable information will serve as a foundation for our project planning phase, as we chart the course to develop our AI-driven incident response platform with a well-informed strategy and timeline."

Stage – 2

Overview: -

An overview of a Nessus scan typically involves the following key steps and aspects:

This project focuses on the practical evaluation and security analysis of the proposed context-aware access control system. After defining the Zero Trust architecture in Stage-1, this phase concentrates on identifying potential security weaknesses and validating how effectively the system protects applications and network resources. The objective is to ensure that the implemented access control policies properly verify users, devices, and access conditions before granting connectivity to internal applications.

➢ In this stage, vulnerability assessment tools and security testing techniques are used to analyze the target environment. Security scanning tools such as Nessus and web application testing methods are applied to detect vulnerabilities including injection flaws, authentication weaknesses, misconfigurations, and access control issues. These tools simulate potential attack scenarios and generate reports that highlight security risks within the system. Each detected vulnerability is carefully examined to understand its cause, severity level, and possible impact on the application or organizational infrastructure.

➢ The analysis performed in this stage also helps demonstrate how context-aware access policies improve security compared to traditional network access models. By evaluating factors such as user identity verification, device security posture, location awareness, and application sensitivity, the system ensures that only authorized users are allowed to access specific resources. Any attempt to access the system from an untrusted device or suspicious location can be blocked automatically through policy enforcement.

➢ Additionally, the results obtained from vulnerability scanning provide valuable insights for strengthening the system's security posture. The findings help in recommending appropriate mitigation techniques such as input validation, secure authentication mechanisms, encryption protocols, and proper configuration management. These measures significantly reduce the attack surface and protect sensitive data from unauthorized access.

➢ Overall, Stage-2 plays a critical role in validating the effectiveness of the implemented Zero Trust access model. By performing vulnerability assessment and analyzing potential threats, this stage ensures that the system remains secure, reliable, and capable of protecting organizational applications frommoderncyber-attacks.

S. No	Vulnerability Name	Severity	Plugins
1.	Cross-site scripting (stored)	Low	Web application Scanner
2.	Broken Access Control	Medium	SSH weak HMAC Algorithms Supported
3.	SQL Injection	Medium	N/A
4.	Broken Authentication	Low	HSTS Missing from Server
5.	Insecure Direct Object Reference	Info	Service Detection
6.	Security Misconfiguration	Low	SSL cipher Block Chaining cipher suites
7.	Cross-Site Request Forgery	Low	HTTP Form- Based Authentication Detected

Target Website: testfire.net
Target IP Address: 54.83.41.36

Report:

1. **VULNERABILITY NAME**: Cross site scripting (stored)

   SEVERITY: Low

   PLUGIN: web application scanner

   PORT: N/A

   DESCRIPTION: It occurs when a malicious script is injected directly into a vulnerable web application. Reflected XSS involves the reflection of a malicious script of a web application, onto a user's browser.

   SOLUTION:
   - When interacting with databases, use parameterized statements or prepared statements to avoid SQL injection, which can be a vector for XSS.
   - Sanitize and validate all user inputs and ensure that any data displayed on the web page properly encoded. Use output encoding libraries or functions to prevent script injection.

   BUSINESS IMPACT: The application stores dangerous data in a database, message forum, visitor log, or other trusted data store. Later, the dangerous data is subsequently read back into the application and included in dynamic content. From an attacker's perspective, the optimal place to inject malicious content is in an area that is displayed to either many users or particularly interesting users. Interesting users typically have elevated privileges in the application or interact with sensitive data that is valuable to the attacker. If one of these users executes malicious content, the attacker may be able to perform privileged operations on behalf of the user or gain access to sensitive data belonging to the user.

2. VULNERABILITY NAME: Broken access control

   SEVERITY: medium

   PLUGIN: SSH weak HMAC Algorithms supported

PORT: N/A

DESCRIPTION: When access control checks are not applied consistently - or not at all - users are able to access data or perform actions that they should not be allowed to perform. This can lead to a wide range of problems, including information exposures, denial of service, and arbitrary code execution.

SOLUTION:

- Ensure that the application has a robust authentication and authorization mechanism in place. Users should be authenticated before any sensitive actions are performed, and authorization checks should be conducted to verify that users have the necessary permissions for the requested operation.
- Ensure that session management is secure. Use strong session IDs, enforce session timeouts, and regenerate session tokens after login. Implement session fixation protection to prevent session hijacking.

BUSINESS IMPACT: wide-ranging business impact, including financial losses, reputation damage, legal consequences, operational disruption, and loss of customer trust. Addressing this vulnerability is crucial to protect the organization and its stakeholders from these potential negative effects.

3. VULNERABILITY NAME: SQL injection

   SEVERITY: medium

   PLUGIN: N/A (typically would be a security testing tool)

   PORT: port 80/443

   DESCRIPTION: The product constructs all or part of an SQL command using externally influenced input from an upstream component, but it does not neutralize or incorrectly neutralizes special elements that could modify the intended SQL command when it is sent to a downstream component.

   SOLUTION:

   - If parameterized statements are not feasible, use proper input validation and escaping mechanisms to sanitize user inputs before they are used in SQL queries. This helps prevent malicious code injection.
   - Utilize stored procedures for database operations. This can help separate SQL code from application code and reduce the risk of SQL injection.
   - Implement strict input validation for all user inputs, ensuring that data adheres to the expected format and structure.

   BUSINESS IMPACT: In summary, it is crucial to underscore that CWE-89, known as SQL Injection, can exert a profound and diverse business impact. This encompasses critical facets such as data breaches, financial setbacks, harm to reputation, legal ramifications, and operational turmoil. Hence, the imperative of preventing and remedying SQL injection vulnerabilities cannot be overstated, as it is indispensable for fortifying the security and continuity of an organization's applications and data.

4. VULNERABILITY NAME: Broken authentication
   SEVERITY: Low

PLUGIN: HSTS Missing from HTTPS Server (plugin id:142960)
PORT: port 443

DESCRIPTION: The system's authorization functionality does not prevent one user from gaining access to another user's data or record by modifying the key value identifying the data.

SOLUTION:

- Implement strong authentication mechanisms, such as multi-factor authentication (MFA) or two-factor authentication (2FA), to ensure that only authorized users can access the system.
- Implement proper authorization controls to ensure that users have the necessary permissions to access specific functions or data. Use role-based access control (RBAC) to manage user privileges effectively.
- Deploy intrusion detection and prevention systems (IDS/IPS) to monitor and block suspicious activities related to authentication and authorization.

BUSINESS IMPACT: To effectively mitigate the business impact of CWE-285, it is imperative that organizations place a strong emphasis on bolstering their authentication and session management practices. This should encompass the adoption of multi-factor authentication, the secure storage of credentials, meticulous session handling, and a commitment to conducting routine security assessments and testing. The rectification of these vulnerabilities stands as a critical imperative, safeguarding sensitive data, user identities, and the organization's overarching security stature and reputation.

5. VULNERABILITY NAME: Insecure Direct object Reference

SEVERITY: Info

PLUGIN: Service Detection (plugin id:22964)

PORT: port 80/443

DESCRIPTION: Insecure Direct Object Reference (IDOR) is a vulnerability that arises when attackers can access or modify objects by manipulating identifiers used in a web application's URLs or parameters. It occurs due to missing access control checks, which fail to verify whether a user should be allowed to access specific data.

SOLUTION:

- Utilize parameterized queries or prepared statements when interacting with databases. This prevents user-controlled input from being executed as SQL code.
- Implement a Content Security Policy to mitigate the risk of cross-site scripting (XSS) attacks, which can be used to execute unauthorized queries.
- Implement account lockout mechanisms that temporarily suspend accounts after a specified number of unauthorized access attempts to prevent brute force attacks.

BUSINESS IMPACT: To effectively mitigate the business impact of CWE-639, organizations must make it a top priority to fortify their access control and authorization mechanisms. This entails the implementation of robust security measures, the regular conduct of comprehensive security assessments, and the deployment of intrusion detection systems to promptly identify and counter unauthorized access attempts. These measures stand as critical in the defence of sensitive data, the overall security of systems, and the preservation of the organization's esteemed reputation.

6. VULNERABILITY NAME: Security Misconfiguration

SEVERITY: low

PLUGIN: SSL cipher Block chaining cipher suites

PORT: port 443

DESCRIPTION: The product specifies permissions for a security-critical resource in a way that allows that resource to be read or modified by unintended actors.

SOLUTION:

- Implement a change management process to document and review all changes to system configurations. This helps prevent unauthorized or unintended alterations.
- Follow security guidelines provided by software and hardware vendors to ensure secure configurations.
- Regularly perform compliance checks against industry standards and regulations to ensure that configurations remain in compliance with security requirements.

BUSINESS IMPACT: To effectively lessen the business consequences stemming from CWE-732, organizations must place a strong emphasis on bolstering their configuration management practices. This involves the establishment of secure default settings, the routine conduct of comprehensive security evaluations, and the deployment of intrusion detection systems to promptly detect and address misconfigurations and vulnerabilities. These actions are of paramount importance in safeguarding sensitive data, fortifying system security, and preserving the organization's esteemed reputation.

7. VULNERABILITY NAME: Cross-Site Request Forgery

SEVERITY: Low

PLUGIN: HTTP Form-Based Authentication Detected

PORT: port 8080/443

DESCRIPTION: Cross-Site Request Forgery (CSRF) is an attack that forces an end user to execute unwanted actions on a web application in which they're currently authenticated. With a little help of social engineering (such as sending a link via email or chat), an attacker may trick the users of a web application into executing actions of the attacker's choosing. If the victim is a normal user, a successful CSRF attack can force the user to perform state changing requests like transferring funds, changing their email address, and so forth. If the victim is an administrative account, CSRF can compromise the entire web application.

SOLUTION:

- Implement Synchronizer Token Patterns to include unique, randomly generated tokens in each HTTP request and validate these tokens on the server side to verify the authenticity of the request.
- Utilize the Same Site attribute in cookies to restrict their usage to same-site requests only, preventing them from being sent along with cross-site requests and effectively mitigating CSRF attacks.

- Utilize framework-specific protections and security features to prevent CSRF attacks. Many modern web frameworks have built-in mechanisms and libraries for handling CSRF vulnerabilities effectively.
- Implement a Content Security Policy to restrict the sources from which various types of content can be loaded. By specifying the trusted sources of content, you can minimize the risk of malicious code execution and reduce the likelihood of successful CSRF attacks.

BUSINESS IMPACT: To effectively reduce the business repercussions of CWE-352, organizations must prioritize secure coding practices, including the incorporation of anti-CSRF tokens, while also conducting routine security assessments and testing to detect and rectify CSRF vulnerabilities. Furthermore, educating users on secure browsing practices is pivotal in preventing CSRF attacks. These actions are of paramount importance in the protection of data and in upholding the trust of customers and partners.

8. VULNERABILITY NAME: Cleartext Transmission of Sensitive Information

SEVERITY: Medium

PLUGIN: Cleartext Transmission of Sensitive Information over HTTP

PORT: port 443

DESCRIPTION: The product transmits sensitive or security-critical data in cleartext in a communication channel that can be sniffed by unauthorized actors. Many communication channels can be "sniffed" (monitored) by adversaries during data transmission. For example, in networking, packets can traverse many intermediary nodes from the source to the destination, whether across the internet, an internal network, the cloud, etc. Some actors might have privileged access to a network interface or any link along the channel, such as a router, but they might not be authorized to collect the underlying data. As a result, network traffic could be sniffed by adversaries, spilling security-critical data.

SOLUTION:

- Educate users and employees on secure data handling practices, such as recognizing secure websites (look for "https") and avoiding insecure Wi-Fi networks.
- Employ data masking or redaction techniques to replace sensitive data with placeholders or cryptographic representations during transmission.
- Implement encryption for data in transit. Use secure encryption protocols such as TLS/SSL to protect sensitive information during transmission.
- Use secure communication protocols for transmitting data, such as HTTPS for web applications, and ensure that the selected protocols are kept up to date.

BUSINESS IMPACT: To effectively reduce the business consequences associated with CWE-319, organizations must prioritize the adoption of secure data transmission practices. This includes the utilization of encryption and robust, secure protocols. The routine conduct of security assessments and testing is pivotal in pinpointing and remedying vulnerabilities linked to data transmission. Furthermore, the education of users on secure data handling practices plays a vital role in proactively preventing data exposure incidents. These actions are of paramount importance in the protection of sensitive data and in preserving the trust of both customers and partners.

9. VULNERABILITY NAME: Clickjacking

SEVERITY: Info

PLUGIN: HTML Rendering Layer Vulnerability

PORT: port 80/443

DESCRIPTION: The web application does not restrict or incorrectly restricts frame objects or UI layers that belong to another application or domain, which can lead to user confusion about which interface the user is interacting with.

SOLUTION:

- Set the X-Frame-Options HTTP response header to deny or same-origin to control how your site can be framed. This is supported by most modern browsers.
- Utilize frame-busting JavaScript code in web applications to prevent the embedding of your site within malicious frame's. This code can disrupt clickjacking attempts.
- Implement a Content Security Policy to restrict which domains can embed your site in iframes. This can help prevent unauthorized framing.
- Implement additional security controls to prevent UI redress attacks, such as clickjacking, within your web application.

BUSINESS IMPACT: To effectively reduce the business consequences resulting from CWE-1021, organizations must prioritize the implementation of protective measures, such as frame-busting code. Simultaneously, educating users on safe browsing practices plays a crucial role in preventing clickjacking incidents. Additionally, routine security assessments and testing are pivotal for identifying and mitigating vulnerabilities associated with clickjacking. These actions are of paramount importance in upholding user trust, ensuring data protection, and safeguarding the organization's reputation.

10. VULNERABILITY NAME: web server allows password auto-completion

SEVERITY: Low

PLUGIN: SSL/TLS Diffie-Hellman Modulus <= 1024 Bits (Logjam)

PORT: port 443

DESCRIPTION: Weaknesses in this category are related to the design and implementation of data confidentiality and integrity. Frequently these deal with the use of encoding techniques, encryption libraries, and hashing algorithms. The weaknesses in this category could lead to a degradation of the quality data if they are not addressed.

SOLUTION:

- Implement secure password storage mechanisms, such as using strong and salted cryptographic hashing algorithms like crypt or script.
- Implement robust encryption key management practices, including secure key storage, key rotation, and access controls.
- Utilize well-established and proven cryptographic libraries and algorithms for encryption and decryption.

- Use secure encryption protocols, like TLS/SSL, for data transmission over networks to protect data in transit

BUSINESS IMPACT: To effectively lessen the business consequences associated with CWE-310, organizations must prioritize the adoption of secure cryptographic practices. This entails ensuring proper password storage and robust encryption key management. Simultaneously, conducting routine security assessments and testing is pivotal in detecting and mitigating vulnerabilities related to cryptographic issues. Striving for compliance with pertinent data protection regulations and industry standards is equally essential. These actions are of paramount importance in the protection of sensitive data, the upholding of user trust, and the preservation of the organization's esteemed reputation.

S. NO	VULNERABILITY NAME	SEVERITY	PLUGINS
1.	Cross-site scripting (DOM based XSS)	Medium	Multiple XSS
2.	jQuery 1.3<3.5.0 Multiple XSS	Medium	CGI Abuse: XSS
3.	Information exposure (web server HTTP header internal IP disclosure)	Low	Webserver HTTP Header Internal IP Disclosure
4.	Information exposure (SSL (Multiple Issues))	Info	SSL Certificate Information
5.	Information exposure (HTTP (Multiple Issues))	Info	HTTP Information
6.	OS identification failed	Info	TCP/IP Remote Probing

TARGET WEBSITE: vit.ac.in

TARGET IP ADDRESS: 122.184.65.22

Report:

1. VULNERABILITY NAME: Cross-site scripting (DOM based XSS)

   SEVERITY: Medium

   PLUGIN: Multiple XSS

   PORT: port 443

   DESCRIPTION: It occurs when a malicious script is injected directly into a vulnerable web application. Reflected XSS involves the reflection of a malicious script of a web application, onto a user's browser. DOM-XSS: DOM stands for Document Object Model and is a programming interface for HTML and XML documents. It represents that programs can change the document structure, style and content. A webpage is a document, and this document can be either displayed in the browser window or as the HTML Source.

SOLUTION:

- When interacting with databases, use parameterized statements or prepared statements to avoid SQL injection, which can be a vector for XSS.
- Sanitize and validate all user inputs and ensure that any data displayed on the web page is properly encoded. Use output encoding libraries or functions to prevent script injection.
- Avoid using DOM manipulation methods that can introduce vulnerabilities. Be cautious with inner HTML, document. Write, and other methods that can execute scripts.
- Consider using well-established JavaScript frameworks and libraries that include built-in security features to prevent DOM XSS.

BUSINESS IMPACT: The application stores dangerous data in a database, message forum, visitor log, or other trusted data store. Later, the dangerous data is subsequently read back into the application and included in dynamic content. From an attacker's perspective, the optimal place to inject malicious content is in an area displayed to many users or particularly interesting users. Interesting users typically have elevated privileges in the application or interact with sensitive data that is valuable to the attacker. If one of these users executes malicious content, the attacker may be able to perform privileged operations on behalf of the user or gain access to sensitive data belonging to the user.

2. VULNERABILITY NAME: jQuery 1.2 < 3.5.0 Multiple XSS

   SEVERITY: Medium

   PLUGIN: CGI Abuse: XSS

   PORT: port 443

   DESCRIPTION: It occurs when a malicious script is injected directly into a vulnerable web application. Reflected XSS involves the reflection of a malicious script of a web application, onto a user's browser. Stored XSS (Persistent XSS): In this type of attack, the malicious script is permanently stored on a server or in a database, often in user-generated content like comments or forum posts. When other users view the affected content, the script executes in their browsers, potentially compromising their data or sessions. Reflected XSS: Reflected XSS occurs when the malicious script is embedded in a URL and immediately executed when a victim clicks on a manipulated link. The script is not stored on a server but is reflected off a web application, making it a one-time attack. DOM-based XSS: DOM stands for Document Object Model and is a programming interface for HTML and XML documents. It represents that programs can change the document structure, style and content. A webpage is a document, and this document can be either displayed in the browser window or as the HTML Source.

   SOLUTION:

   - The most important recommendation is to update to the latest version of jQuery. Newer versions typically include security patches, bug fixes, and performance improvements.
   - Stay vigilant about jQuery updates and security announcements. Subscribe to relevant security mailing lists or forums to receive notifications about new releases.
   - A Content Security Policy (CSP) can help mitigate various types of client-side vulnerabilities, including Cross-Site Scripting (XSS). Consider implementing a CSP in your web application to restrict the sources from which scripts can be loaded.

- If updating jQuery is not immediately possible, review your code for potential security issues. Ensure that you are escaping or sanitizing user inputs and that you are using safe coding practices to prevent XSS and other security vulnerabilities.

BUSINESS IMPACT: The application stores dangerous data in a database, message forum, visitor log, or other trusted data store. Later, the dangerous data is subsequently read back into the application and included in dynamic content. From an attacker's perspective, the optimal place to inject malicious content is in an area displayed to many users or particularly interesting users. Interesting users typically have elevated privileges in the application or interact with sensitive data that is valuable to the attacker. If one of these users executes malicious content, the attacker may be able to perform privileged operations on behalf of the user or gain access to sensitive data belonging to the user.

3. VULNERABILITY NAME: Information Exposure (Web Server HTTP Header Internal IP Disclosure)

   SEVERITY: Low

   PLUGIN: Web Server HTTP Header Internal IP Disclosure

   PORT: port 80

   DESCRIPTION: The product exposes sensitive information to an actor that is not explicitly authorized to have access to that information.

   SOLUTION:
   - Carefully review your web server's configuration files (e.g., Apache's httpd.conf, Nginx's nginx.conf) to identify anywhere internal IP addresses are inadvertently included in HTTP response headers.
   - Remove or replace any internal IP addresses in the server configuration files with appropriate placeholders or public IP addresses.
   - If you use proxy servers or load balancers, ensure that they are configured to hide internal IP addresses and only expose public IP addresses instances in HTTP headers.
   - Review and configure HTTP response headers (e.g., Server, X-Powered-By) to limit the exposure of server and technology details, reducing the potential for disclosing internal IP addresses.

   BUSINESS IMPACT: To effectively reduce the business consequences arising from SSL issues, organizations must give utmost priority to establishing a secure SSL/TLS configuration, conducting frequent security assessments and testing. Adhering to data protection regulations and industry standards is imperative. These actions play a pivotal role in safeguarding sensitive data, upholding user trust, and safeguarding the organization's esteemed reputation.

4. VULNERABILITY NAME: Information Exposure (SSL (Multiple Issues))

   SEVERITY: Info

   PLUGIN: SSL certificate Information

   PORT: port 443

DESCRIPTION: The product exposes sensitive information to an actor that is not explicitly authorized to have access to that information.
SOLUTION:

- Configure your web server to use the latest and strongest encryption protocols, such as TLS 1.2 or higher. Disable outdated and insecure protocols like SSLv3.
- Implement Perfect Forward Secrecy to ensure that even if the private key is compromised, previously recorded encrypted traffic remains secure.
- Safeguard your private keys by storing them securely and ensuring they are not accessible to unauthorized parties.

BUSINESS IMPACT: To effectively reduce the business consequences arising from SSL issues, organizations must give utmost priority to establishing a secure SSL/TLS configuration, conducting frequent security assessments and testing. Adhering to data protection regulations and industry standards is imperative. These actions play a pivotal role in safeguarding sensitive data, upholding user trust, and safeguarding the organization's esteemed reputation.

5. VULNERABILITY NAME: Information Exposure (HTTP (Multiple Issues))

   SEVERITY: Info

   PLUGIN: HTTP information

   PORT: port 80 /443

   DESCRIPTION: The product exposes sensitive information to an actor that is not explicitly authorized to have access to that information

   SOLUTION:

   - Minimize the use of unnecessary redirects as they can increase page load times. Use 301 (permanent) redirects for consistent content.
   - Regularly check for broken links (404 errors) and fix them to ensure users find the content they are looking for.
   - Properly configure CORS to control which domains are allowed to make requests to your server.
   - Migrate your website from HTTP to HTTPS to ensure secure data transmission. This is essential for user trust and search engine ranking.

   BUSINESS IMPACT: To effectively reduce the business repercussions of HTTP issues, organizations must place a strong emphasis on ongoing monitoring and enhancement of their web services. Consistent performance testing and security assessments, coupled with proactive maintenance, are indispensable to ensure a seamless and secure online presence. Furthermore, offering timely and helpful customer support and transparent communication when HTTP issues arise is instrumental in mitigating adverse effects on user trust and satisfaction.

6. VULNERABILITY NAME: OS Identification Failed

   SEVERITY: Info

PLUGIN: TCP/IP remote probing

PORT: N/A

DESCRIPTION: OS identification failed refers to the inability of a system or software to accurately recognize the operating system running on a device or network. This issue can result from various factors such as misconfigurations, network issues, or compatibility problems. The failure to identify the OS correctly can lead to security vulnerabilities, hindered software updates, and potential operational disruptions. Resolving this issue promptly is crucial to ensure accurate security measures and the smooth functioning of the system or network.

SOLUTION:

- Minimize the use of unnecessary redirects as they can increase page load times. Use 301 (permanent) redirects for consistent content.
- Regularly check for broken links (404 errors) and fix them to ensure users find the content they are looking for.
- Properly configure CORS to control which domains are allowed to make requests to your server.
- Migrate your website from HTTP to HTTPS to ensure secure data transmission. This is essential for user trust and search engine ranking.

BUSINESS IMPACT: It may lead to operational inefficiencies, potential security vulnerabilities, non-compliance with industry regulations, and a possible loss of customer trust. Such failures could result in downtime, compromised data security, regulatory penalties, and damage to the company's reputation, potentially leading to financial losses and decreased market competitiveness. Resolving this issue promptly is crucial to mitigate these risks and maintain the smooth functioning of the organization.

Stage – 3

Title:

Design and apply context-aware access controls based on user identity, device posture, location, and application risk.

Overview

Context-aware access control is an important component of a Zero Trust security architecture. Instead of granting access based only on username and password, this approach evaluates multiple factors before allowing a user to access an application. These factors include user identity, device posture, location, and the sensitivity of the application being accessed. By analyzing these elements in real time, the system can decide whether to allow, deny, or limit access. This ensures that security policies are dynamic and responsive to potential risks, thereby protecting enterprise resources more effectively.

Identity

User identity is the foundation of context-aware access control. Authentication is typically performed through an Identity Provider (IdP) such as Azure Active Directory or Okta using protocols like SAML or OAuth. When a user logs in through systems like Zscaler Private Access (ZPA), the identity provider verifies the user using multi-factor authentication (MFA). Along with authentication, the system retrieves user attributes such as department, role, and group membership. These attributes help create detailed access policies. For example, only members of the Finance department may be allowed to access payroll applications, ensuring that access is granted strictly according to organizational roles.

Device Posture

Device posture refers to the security condition of the device used to access the application. A secure device reduces the risk of unauthorized access or malware attacks. Systems like ZPA can integrate with endpoint security tools and device management platforms to check device compliance. The system evaluates factors such as whether antivirus software is active, disk encryption is enabled, the firewall is running, and the operating system is updated with the latest security patches. If the device fails these checks, access may be denied or restricted until the device meets the required security standards.

Application Risk

Applications within an organization vary in sensitivity and importance. Some applications, such as internal information portals, may be considered low risk, while others like financial systems, customer databases, or source code repositories are highly sensitive. Context-aware access control categorizes applications based on their risk level. High-risk applications require stricter security policies, such as stronger authentication methods, restricted access locations, and additional monitoring. By assigning risk levels to applications, organizations can apply appropriate security controls and protect critical resources effectively.

Context-Aware Access Control

Context-aware access control combines multiple factors—identity, device posture, location, and application risk—to make intelligent access decisions. Instead of applying a single static rule, the system dynamically evaluates the context of each access request. For example, a user accessing an application from a trusted device and secure network may be granted full access. However, if the same user connects from an unknown location or an insecure device, the system may require additional authentication or limit access privileges. This adaptive

approach ensures that access decisions are based on real-time risk evaluation, supporting the Zero Trust principle of “never trust, always verify.”

Conclusion:

Stage 1

In conclusion, web application testing plays an essential role in ensuring the security, performance, and reliability of modern web-based systems. It involves evaluating multiple aspects of an application such as functionality, usability, performance, and security in order to detect defects and vulnerabilities before they affect users. Effective testing improves the overall quality of applications and strengthens their ability to withstand cyber threats. As web services continue to become a critical part of everyday life, proper testing practices are necessary to provide users with safe, efficient, and reliable digital experiences.

Stage 2

The vulnerability assessment performed using the Nessus scanner provided valuable insights into the security condition of the tested website. The report identified several vulnerabilities and categorized them according to severity levels, with particular emphasis on critical and high-risk issues that require immediate remediation. Suggested mitigation steps include applying software patches, modifying configurations, and implementing stronger security controls. The assessment also considered compliance with industry standards such as OWASP security guidelines. Overall, the report highlights the importance of continuous monitoring, timely updates, and proactive security management to maintain a strong defensive posture against emerging threats.

Stage 3

The deployment of Zscaler Private Access represents a significant transition from traditional network-based security to a modern Zero Trust architecture centered on user identity. By enforcing least-privilege access and hiding internal applications from the public internet, the solution effectively reduces the risk of lateral movement and network-based attacks. This approach strengthens the organization’s overall cybersecurity framework while providing secure and seamless access for remote users. Additionally, it reduces dependence on legacy VPN systems and improves visibility into user and application activities. Ultimately, the implementation creates a scalable and future-ready security environment that supports secure digital transformation.

Future Scope:

Stage 1: Future Scope of Web Application Testing

The future of web application testing is rapidly evolving due to the increasing dependence on web technologies and digital platforms. As organizations continue to expand their online presence, the importance of testing web applications will grow significantly.

Key future developments include:

• **Growing Demand:** As more businesses rely on web-based platforms, the need for comprehensive testing to ensure functionality, reliability, and security will increase.

• **Advanced Technologies:** Emerging technologies such as Artificial Intelligence, Internet of Things (IoT), and blockchain will make web applications more complex, requiring advanced testing methods and specialized tools.

• **Stronger Security Focus:** With the rising number of cyberattacks and data breaches, security testing will become a primary focus to identify vulnerabilities and ensure data protection and regulatory compliance.

• **Mobile and Cross-Browser Compatibility:** Since users access web applications from multiple devices and browsers, ensuring compatibility and responsiveness across platforms will remain a key testing objective.

Stage 2: Future Scope of Vulnerability Testing

The future scope of vulnerability testing in web applications is expanding as digital ecosystems continue to grow and cyber threats become more sophisticated.

Important developments include:

• **AI-Driven Vulnerability Detection:** Artificial Intelligence and Machine Learning will help automate vulnerability detection by analyzing large volumes of code and identifying hidden security weaknesses.

• **IoT and Mobile Security Testing:** As IoT devices and mobile applications become widespread, security testing will extend to protect connected devices and their communication interfaces.

• **Blockchain Security:** With the adoption of blockchain technology and digital currencies, specialized testing methods will be required to ensure secure transactions and data integrity.

• **Serverless Architecture Testing:** As organizations adopt serverless computing models, testing will focus on securing APIs and cloud-based functions from vulnerabilities.

Stage 3: Future Scope of Context-Aware Access Control

The successful implementation of Zscaler Private Access opens opportunities for further enhancement of the organization's cybersecurity infrastructure.

Possible future developments include:

• **Integration with Zscaler Internet Access (ZIA):** Combining ZPA and ZIA to build a unified Secure Access Service Edge (SASE) platform for managing both internal and internet traffic through centralized policies.

• **Advanced Device Security Monitoring:** Incorporating real-time device risk analysis through endpoint security tools to automatically block compromised devices.

• **Privileged Remote Access:** Providing secure and temporary access for third-party vendors and contractors while maintaining strict monitoring and auditing controls.

• **IoT and Operational Technology Segmentation:** Applying Zero Trust principles to IoT devices and industrial systems to restrict communication only to authorized management servers.

• **AI-Based Threat Detection:** Using machine learning to analyze access patterns and detect unusual behavior such as abnormal login times or suspicious locations.

• **Cloud Workload Security:** Protecting communication between containers, microservices, and cloud workloads across multi-cloud environments.

• **Automated Access Management:** Automatically granting or revoking user permissions based on changes in employee roles or employment status.

• **Digital Experience Monitoring:** Implementing monitoring tools to track user experience and identify performance issues before they affect productivity.

Topics Explored:

1. Zscaler
2. Zero Trust Security Model
3. Zscaler Private Access (ZPA)
4. Virtual Private Network (VPN)
5. Malware Information Sharing Platform (MISP)
6. College Network Infrastructure Analysis
7. Deployment of Security Operations Center (SOC) in the College
8. Threat Intelligence
9. Incident Response

Tools Explored:

- **Nessus Vulnerability Scanner** – Used to perform vulnerability assessment and identify security weaknesses in web applications and systems. It scans networks and applications to detect misconfigurations, outdated software, and potential security threats, helping organizations improve their security posture.